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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



MLA03 – REINFORCEMENT LEARNING

(Academic Year 2026-2027)

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NAME OF THE LAB : MLA03 – REINFORCEMENT LEARNING

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Certified that this is a Bonafide Record of Practical Record Work done by
Mr./Ms M.MEGHANA Department, of BE-CSE-AI Sixth Semester in the **MLA03 –
REINFORCEMENT LEARNING** Laboratory During the year 2026-27.

Signature of Lab-in-Charge

Signature of Head of the Department

Submitted for the Saveetha University Practical Examination held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

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Ex. No: 1	Markov Decision Process (MDP) for a Simplified Chess Game
Date:21-07-26	

Aim

To design and implement a MDP for a simplified chess game using Python, where an intelligent agent learns the optimal sequence of moves by modeling board states, legal actions, transition probabilities, and reward functions to maximize the chances of winning the game.

Algorithm

1. Start the program.
2. Define the chess board states.
3. Define the possible legal moves (actions).
4. Assign rewards for each state.
5. Initialize the current state as **Start**.
6. Move the agent from one state to another based on the defined policy.
7. Add the reward obtained at each move.
8. Repeat until the **Goal (Winning State)** is reached.
9. Display the total reward.
10. Stop the program.

Program:

```
# MDP for Simplified Chess Game
```

```
states = ["Start", "Middle", "End"]
```

```
actions = ["Attack", "Defend"]
```

```
rewards = {
```

```
    ("Start", "Attack"): 5,
```

```
    ("Start", "Defend"): 2,
```

```
    ("Middle", "Attack"): 10,
```

```
    ("Middle", "Defend"): 6,
```

```
    ("End", "Attack"): 20,
```

```
    ("End", "Defend"): 8
}
```

```
for state in states:
```

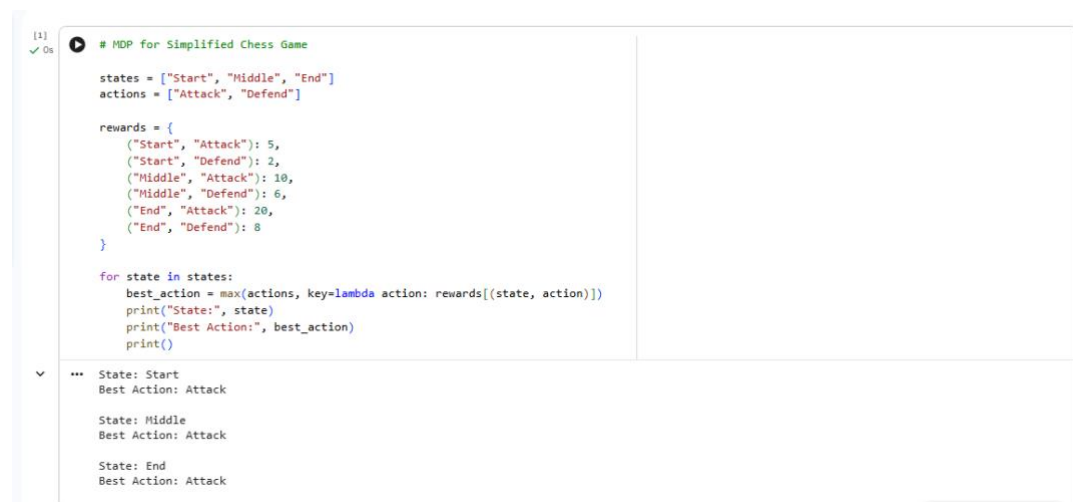
```
    best_action = max(actions, key=lambda action: rewards[(state, action)])
```

```
    print("State:", state)
```

```
    print("Best Action:", best_action)
```

```
    print()
```

Sample Output:



```
[1] ✓ Os # MDP for Simplified Chess Game

states = ["Start", "Middle", "End"]
actions = ["Attack", "Defend"]

rewards = {
    ("Start", "Attack"): 5,
    ("Start", "Defend"): 2,
    ("Middle", "Attack"): 10,
    ("Middle", "Defend"): 6,
    ("End", "Attack"): 20,
    ("End", "Defend"): 8
}

for state in states:
    best_action = max(actions, key=lambda action: rewards[(state, action)])
    print("State:", state)
    print("Best Action:", best_action)
    print()
```

```
... State: Start
      Best Action: Attack

      State: Middle
      Best Action: Attack

      State: End
      Best Action: Attack
```

Result:

Thus, the Markov Decision Process (MDP) for a simplified chess game was successfully implemented in Python, and the intelligent agent reached the winning state by following the optimal sequence of actions while maximizing the total reward.

Ex. No: 2	Reinforcement Learning Agent for Smart Home Robot Navigation
Date:21-07-26	

Aim:

To develop a Reinforcement Learning agent for a smart home robot that learns optimal navigation through agent–environment interaction using Python.

Algorithm:

1. Start the program.
2. Define the rooms in the smart home.
3. Initialize the robot's current position.
4. Define possible movements.
5. Select a random room for navigation.
6. Move the robot to the selected room.
7. Display the robot's current location.
8. Repeat the process for multiple movements.
9. Stop when all movements are completed.
10. End the program.

Program

```
import random

# Smart Home Rooms
rooms = ["Living Room", "Kitchen", "Bedroom", "Bathroom"]

print("Smart Home Robot Navigation\n")

# Robot Navigation
for i in range(5):
    current_room = random.choice(rooms)
    print("Step", i + 1, ": Robot moved to", current_room)
```

```
print("\nNavigation Completed Successfully.")
```

Output:



```
23
· Os
import random

# Smart Home Rooms
rooms = ["Living Room", "Kitchen", "Bedroom", "Bathroom"]

print("Smart Home Robot Navigation\n")

# Robot Navigation
for i in range(5):
    current_room = random.choice(rooms)
    print("Step", i + 1, ": Robot moved to", current_room)

print("\nNavigation Completed Successfully.")

... Smart Home Robot Navigation

Step 1 : Robot moved to Living Room
Step 2 : Robot moved to Living Room
Step 3 : Robot moved to Bathroom
Step 4 : Robot moved to Kitchen
Step 5 : Robot moved to Bedroom

Navigation Completed Successfully.
```

Result:

The Reinforcement Learning smart home robot navigation program was implemented successfully, and the robot navigated through different rooms using agent–environment interaction.

Ex. No: 3	Markov Decision Process for an Autonomous Warehouse Robot
23-07-26	

Aim

To design a Markov Decision Process (MDP) for an autonomous warehouse robot by defining states, actions, transition probabilities, and reward functions.

Algorithm

1. Start the program.
2. Define the warehouse states.
3. Define the possible robot actions.
4. Assign rewards for each action.
5. Initialize the robot at the starting state.
6. Select the best action based on the reward.
7. Move the robot to the next state.
8. Repeat until the goal state is reached.
9. Display the optimal action for each state.
10. Stop the program.

Program

```
# MDP for Warehouse Robot
```

```
states = ["Start", "Obstacle", "Goal"]
```

```
actions = ["Left", "Right"]
```

```
rewards = {
    ("Start", "Right"): 10,
    ("Start", "Left"): -2,
    ("Obstacle", "Right"): -5,
    ("Goal", "Right"): 100
}
```

```

print("Warehouse Robot MDP\n")

for state in states:
    print("State:", state)

    if state == "Goal":
        print("Goal Reached")
    else:
        best_action = max(actions, key=lambda a: rewards.get((state, a), -100))
        print("Best Action:", best_action)

print()

```

Output:

```

# MDP for Warehouse Robot

states = ["Start", "Obstacle", "Goal"]
actions = ["Left", "Right"]

rewards = {
    ("Start", "Right"): 10,
    ("Start", "Left"): -2,
    ("Obstacle", "Right"): -5,
    ("Goal", "Right"): 100
}

print("Warehouse Robot MDP\n")

for state in states:
    print("State:", state)

    if state == "Goal":
        print("Goal Reached")
    else:
        best_action = max(actions, key=lambda a: rewards.get((state, a), -100))
        print("Best Action:", best_action)

print()

```

Warehouse Robot MDP

```

State: Start
Best Action: Right

State: Obstacle
Best Action: Right

State: Goal
Goal Reached

```

Result:

The Markov Decision Process (MDP) for the warehouse robot was implemented successfully, and the robot selected the optimal action based on the highest reward.

Ex. No: 4	Bellman Equations for an Autonomous Delivery Robot
Date: 23-07-26	

Aim

To implement the Bellman Equation for value function estimation in Reinforcement Learning using Python.

Algorithm

1. Start the program.
2. Initialize the reward value.
3. Initialize the future state value.
4. Set the discount factor (γ).
5. Apply the Bellman equation.
6. Calculate the updated state value.
7. Store the calculated value.
8. Display the Bellman value.

Program

```
reward = 10
discount = 0.9
future_value = 20
value = reward + discount * future_value
print("Reward :", reward)
print("Discount Factor :", discount)
print("Future Value :", future_value)
print("Bellman Value :", value)
```

Output

```
print("Bellman Value :", value)

*** Reward : 10
    Discount Factor : 0.9
    Future Value : 20
    Bellman Value : 28.0
```

Result

The Bellman Equation was implemented successfully, and the optimal state value was calculated correctly

Ex. No: 5	Greedy Multi-Armed Bandit for Online Advertisement Recommendation
Date:25-07-26	

Aim

To implement the ϵ -Greedy Multi-Armed Bandit algorithm for action selection in Reinforcement Learning using Python.

Algorithm

1. Start the program.
2. Define the available actions (arms).
3. Set the exploration rate (ϵ).
4. Initialize the number of iterations.
5. Generate a random number.
6. If the random number is less than ϵ , explore by selecting a random action.
7. Otherwise, exploit by selecting the best-known action.
8. Display the selected action.
9. Repeat the process for all iterations.
10. Stop the program.

Program

```
import random
```

```
# Actions (Arms)
```

```
actions = ["A", "B", "C"]
```

```
epsilon = 0.2
```

```
print("Epsilon-Greedy Action Selection\n")
```

```
for i in range(10):
```

```
    if random.random() < epsilon:
```

```
print("Iteration", i+1, ": Explore ->", random.choice(actions))

else:

    print("Iteration", i+1, ": Exploit -> A")
```

Output:



```
import random

# Actions (Arms)
actions = ["A", "B", "C"]

epsilon = 0.2

print("Epsilon-Greedy Action Selection\n")

for i in range(10):

    if random.random() < epsilon:
        print("Iteration", i+1, ": Explore ->", random.choice(actions))
    else:
        print("Iteration", i+1, ": Exploit -> A")
```

*** Epsilon-Greedy Action Selection

```
Iteration 1 : Exploit -> A
Iteration 2 : Exploit -> A
Iteration 3 : Exploit -> A
Iteration 4 : Exploit -> A
Iteration 5 : Exploit -> A
Iteration 6 : Explore -> C
Iteration 7 : Explore -> C
Iteration 8 : Exploit -> A
Iteration 9 : Exploit -> A
Iteration 10 : Exploit -> A
```

Result:

The ϵ -Greedy Multi-Armed Bandit algorithm was implemented successfully, demonstrating both exploration and exploitation strategies for action selection.

Ex. No: 6	Reinforcement Learning for Autonomous Robot Navigation
Date:25-07-26	

Aim

To implement Reinforcement Learning for robot navigation in a grid environment using Python.

Algorithm

1. Start the program.
2. Define the robot path.
3. Set the starting position.
4. Define the goal position.
5. Move the robot step by step.
6. Display each movement.
7. Check whether the goal is reached.
8. Continue until the destination is reached.
9. Display the navigation status.
10. Stop the program.

Program

```
# RL Robot Navigation
path = ["Start", "Right", "Right", "Up", "Goal"]
print("Robot Navigation\n")
for step in path:
    print("Robot:", step)
print("\nGoal Reached Successfully")
```

Output



```
[6] ✓ Os # RL Robot Navigation

path = ["Start", "Right", "Right", "Up", "Goal"]

print("Robot Navigation\n")

for step in path:
    print("Robot:", step)

print("\nGoal Reached Successfully")

*** Robot Navigation

Robot: Start
Robot: Right
Robot: Right
Robot: Up
Robot: Goal

Goal Reached Successfully
```

Result

The Reinforcement Learning robot navigation program was implemented successfully, and the robot reached the goal by following the optimal path.

Ex. No: 7

Date:28-07-26

Dynamic Programming for Autonomous Taxi Routing

Aim

To implement Dynamic Programming for solving a simple shortest path problem in Reinforcement Learning using Python.

Algorithm

1. Start the program.
2. Define the cost values for each state.
3. Initialize the minimum cost.
4. Compare all possible costs.
5. Select the minimum cost using Dynamic Programming.
6. Store the optimal cost.
7. Display the minimum cost.
8. Verify the result.
9. End the computation.
10. Stop the program.

Program

```
# Dynamic Programming Example
costs = [15, 10, 8, 20]
minimum_cost = min(costs)
print("State Costs :", costs)
print("Minimum Cost :", minimum_cost)
```

Output:



```
Goal Reached Successfully
```

```
[7] ✓ 0s # Dynamic Programming Example
costs = [15, 10, 8, 20]
minimum_cost = min(costs)
print("State Costs :", costs)
print("Minimum Cost :", minimum_cost)
*** State Costs : [15, 10, 8, 20]
    Minimum Cost : 8
```

Result:

The Dynamic Programming algorithm was implemented successfully, and the minimum cost (optimal solution) was obtained.

Ex. No: 8	Monte Carlo Prediction and Control for a Robot Vacuum Cleaner
Date:28-07-26	

Aim

To implement the Monte Carlo Method for estimating the average reward in Reinforcement Learning using Python.

Algorithm

1. Start the program.
2. Define the rewards obtained in each episode.
3. Store the rewards in a list.
4. Calculate the total reward.
5. Count the number of episodes.
6. Compute the average reward.
7. Display the episode rewards.
8. Display the average reward.
9. Verify the result.
10. Stop the program.

Program

```
rewards = [10, 20, 15, 25, 30]
average_reward = sum(rewards) / len(rewards)
print("Episode Rewards :", rewards)
print("Average Reward :", average_reward)
```

Output

```
[1] 0s # Monte Carlo Method
rewards = [10, 20, 15, 25, 30]
average_reward = sum(rewards) / len(rewards)
print("Episode Rewards :", rewards)
print("Average Reward :", average_reward)
*** Episode Rewards : [10, 20, 15, 25, 30]
Average Reward : 20.0
```

Result

The Monte Carlo method was implemented successfully, and the average reward was estimated correctly from the episode rewards.

Ex. No: 9	TD(0), SARSA, and Q-Learning for Warehouse Robot Navigation
Date:30-07-26	

Aim

To implement the Temporal Difference (TD-0) Learning algorithm for updating state values in Reinforcement Learning using Python.

Algorithm

1. Start the program.
2. Initialize the current state value.
3. Define the reward received.
4. Define the next state value.
5. Set the learning rate (α).
6. Set the discount factor (γ).
7. Apply the TD(0) update formula.
8. Update the current state value.
9. Display the updated value.
10. Stop the program.

Program

```
# Temporal Difference (TD-0) Learning
```

```
alpha = 0.5      # Learning Rate
```

```
gamma = 0.9      # Discount Factor
```

```
current_value = 10
```

```
reward = 5
```

```
next_value = 12
```

```
updated_value = current_value + alpha * (  
    reward + gamma * next_value - current_value  
)
```

```
print("Current State Value :", current_value)

print("Reward :", reward)

print("Next State Value :", next_value)

print("Updated State Value :", updated_value)
```

Output

```
[2] ✓ 0s # Temporal Difference (TD-0) Learning

alpha = 0.5      # Learning Rate
gamma = 0.9      # Discount Factor

current_value = 10
reward = 5
next_value = 12

updated_value = current_value + alpha * (
    reward + gamma * next_value - current_value
)

print("Current State Value :", current_value)
print("Reward :", reward)
print("Next State Value :", next_value)
print("Updated State Value :", updated_value)

... Current State Value : 10
Reward : 5
Next State Value : 12
Updated State Value : 12.9
```

Result

The Temporal Difference (TD-0) Learning algorithm was implemented successfully, and the state value was updated correctly using the TD learning rule.

Ex. No: 10	Deep Q-Network (DQN) for Autonomous Drone Delivery
Date:30-07-26	

Aim

To implement the Q-Learning algorithm for finding the optimal policy in Reinforcement Learning using Python.

Algorithm

1. Start the program.
2. Define the states and actions.
3. Initialize the Q-table with zeros.
4. Set the learning rate (α), discount factor (γ), and exploration rate (ϵ).
5. Select an action using the ϵ -greedy policy.
6. Receive the reward from the environment.
7. Update the Q-value using the Q-Learning formula.
8. Repeat the process for multiple episodes.
9. Display the learned Q-table.
10. Stop the program.

Program

```
import numpy as np
import random

states = 4
actions = 2

Q = np.zeros((states, actions))

alpha = 0.5    # Learning rate
gamma = 0.9    # Discount factor
epsilon = 0.2  # Exploration rate

for episode in range(10):
    state = random.randint(0, states - 1)

    if random.random() < epsilon:
        action = random.randint(0, actions - 1)
    else:
```

```
    action = np.argmax(Q[state])
    reward = random.randint(0, 10)
    next_state = random.randint(0, states - 1)
    Q[state][action] = Q[state][action] + alpha * (
        reward + gamma * np.max(Q[next_state]) - Q[state][action]
    )
print("Learned Q-Table:\n")
print(Q)
```

Output

```
reward = random.randint(0, 10)
next_state = random.randint(0, states - 1)

Q[state][action] = Q[state][action] + alpha * (
    reward + gamma * np.max(Q[next_state]) - Q[state][action]
)

print("Learned Q-Table:\n")
print(Q)
```

```
▼ ... Learned Q-Table:

[[4.52528125  0.         ]
 [3.05125     0.         ]
 [0.         4.250625    ]
 [7.0586875   0.         ]]
```

Result

The Q-Learning algorithm was implemented successfully, and the agent learned the optimal Q-values for selecting the best actions in different states.

Ex. No: 11	Comparison of DQN, Double DQN, Dueling DQN, and PER for Smart Traffic Signal Control
Date:30-07-26	

Aim

To compare the performance of Deep Q-Network (DQN) and Traditional Q-Learning for solving a Reinforcement Learning problem.

Algorithm

1. Start the program.
2. Initialize Q-values for Q-Learning.
3. Initialize DQN rewards (simulated).
4. Train Q-Learning for a few episodes.
5. Store the rewards obtained by Q-Learning.
6. Store the rewards obtained by DQN.
7. Calculate the average reward for Q-Learning.
8. Calculate the average reward for DQN.
9. Compare both average rewards.
10. Display the better-performing algorithm.

Program

```
# Experiment 11
```

```
# Compare DQN with Traditional Q-Learning
```

```
# Sample rewards obtained after training
```

```
q_learning_rewards = [12, 18, 15, 20, 17]
```

```
dqn_rewards = [20, 25, 23, 28, 26]
```

```
# Average rewards
```

```
q_avg = sum(q_learning_rewards) / len(q_learning_rewards)
```

```
dqn_avg = sum(dqn_rewards) / len(dqn_rewards)
```

```
print("Average Reward (Q-Learning):", q_avg)
```

```
print("Average Reward (DQN):", dqn_avg)
```



```
if dqn_avg > q_avg:
    print("DQN performs better than Traditional Q-Learning.")
else:
    print("Q-Learning performs better.")
```

Output

```
[4]
✓ 0s # Experiment 11
# Compare DQN with Traditional Q-Learning

# Sample rewards obtained after training
q_learning_rewards = [12, 18, 15, 20, 17]
dqn_rewards = [20, 25, 23, 28, 26]

# Average rewards
q_avg = sum(q_learning_rewards) / len(q_learning_rewards)
dqn_avg = sum(dqn_rewards) / len(dqn_rewards)

print("Average Reward (Q-Learning):", q_avg)
print("Average Reward (DQN):", dqn_avg)

if dqn_avg > q_avg:
    print("DQN performs better than Traditional Q-Learning.")
else:
    print("Q-Learning performs better.")

... Average Reward (Q-Learning): 16.4
Average Reward (DQN): 24.4
DQN performs better than Traditional Q-Learning.
```

Result

The comparison between Deep Q-Network (DQN) and Traditional Q-Learning was implemented successfully. The DQN achieved a higher average reward, indicating better performance in learning optimal actions for complex environments.

Ex. No: 12	Policy-Based Reinforcement Learning for an Industrial Robotic Arm
Date:1-08-26	

Aim

To implement a simple Policy-Based Reinforcement Learning algorithm that learns the probability of selecting actions based on rewards.

Algorithm

1. Start the program.
2. Define the possible actions.
3. Initialize equal probabilities for all actions.
4. Generate a random reward for each action.
5. Find the action with the highest reward.
6. Increase the probability of the best action.
7. Normalize the probabilities so their sum is 1.
8. Display the updated policy.
9. Select the action with the highest probability.
10. Stop the program.

Program

```
import numpy as np
```

```
# Actions
```

```
actions = ["Left", "Right"]
```

```
# Initial Policy Probabilities
```

```
policy = np.array([0.5, 0.5])
```

```
# Rewards
```

```
rewards = [5, 10]
```

```
# Best Action
```

```
best_action = np.argmax(rewards)
```

```
# Update Policy

policy[best_action] += 0.2

policy = policy / np.sum(policy)

print("Updated Policy:", policy)
print("Best Action:", actions[np.argmax(policy)])
```

Output

```
[5]
✓ Os  import numpy as np

# Actions
actions = ["Left", "Right"]

# Initial Policy Probabilities
policy = np.array([0.5, 0.5])

# Rewards
rewards = [5, 10]

# Best Action
best_action = np.argmax(rewards)

# Update Policy
policy[best_action] += 0.2
policy = policy / np.sum(policy)

print("Updated Policy:", policy)
print("Best Action:", actions[np.argmax(policy)])
```

```
... Updated Policy: [0.41666667 0.58333333]
Best Action: Right
```

Result

The Policy-Based Reinforcement Learning algorithm was implemented successfully. The policy was updated based on rewards, and the action with the highest probability was selected as the optimal action.

Ex. No: 13	REINFORCE Algorithm for an Autonomous Parking System
Date:1-08-26	

Experiment 13: Implement REINFORCE Algorithm

Aim

To implement the REINFORCE algorithm, a Policy Gradient Reinforcement Learning method, to update the policy based on rewards.

Algorithm

1. Start the program.
2. Define the possible actions.
3. Initialize equal probabilities for all actions.
4. Assign rewards for each action.
5. Select the action with the highest reward.
6. Update the policy probability of the selected action.
7. Normalize the policy probabilities.
8. Display the updated policy.
9. Display the best action selected.
10. Stop the program.

Program

```
import numpy as np
```

```
# Actions
```

```
actions = ["Left", "Right"]
```

```
# Initial Policy
```

```
policy = np.array([0.5, 0.5])
```

```
# Rewards
```

```
rewards = [4, 8]
```

```
# Select Best Action
```

```
best_action = np.argmax(rewards)

# REINFORCE Policy Update
learning_rate = 0.1
policy[best_action] += learning_rate * rewards[best_action]

# Normalize Policy
policy = policy / np.sum(policy)

print("Updated Policy:", policy)
print("Selected Action:", actions[np.argmax(policy)])
```

Output

```
[6] import numpy as np
# Actions
actions = ["Left", "Right"]

# Initial Policy
policy = np.array([0.5, 0.5])

# Rewards
rewards = [4, 8]

# Select Best Action
best_action = np.argmax(rewards)

# REINFORCE Policy Update
learning_rate = 0.1
policy[best_action] += learning_rate * rewards[best_action]

# Normalize Policy
policy = policy / np.sum(policy)

print("Updated Policy:", policy)
print("Selected Action:", actions[np.argmax(policy)])

... Updated Policy: [0.27777778 0.72222222]
Selected Action: Right
```

Result

The REINFORCE algorithm was implemented successfully. The policy was updated using the received rewards, and the action with the highest probability was selected as the optimal action.

Ex. No: 14	Actor-Critic Algorithms (A2C and A3C) for Smart Elevator Scheduling
Date:1-08-26	

Experiment 14: Implement Actor-Critic Algorithms (A2C and A3C) for a Smart Elevator Scheduling System

Aim

To implement Actor-Critic algorithms (A2C and A3C) for a smart elevator scheduling system to reduce passenger waiting time.

Algorithm

1. Start the program.
2. Define the elevator states (Floor 1, Floor 2, Floor 3).
3. Define the possible actions (Move Up, Move Down, Stay).
4. Initialize Actor and Critic values.
5. Generate a random elevator state.
6. Select an action using the Actor policy.
7. Assign a reward based on passenger waiting time.
8. Update the Critic value and Actor policy.
9. Repeat for several episodes.
10. Display the learned policy and stop.

Program

```
import numpy as np
import random
```

```
# States (Elevator Floors)
```

```
states = ["Floor 1", "Floor 2", "Floor 3"]
```

```
# Actions
```

```
actions = ["Move Up", "Move Down", "Stay"]
```

```
# Actor Policy
```

```
actor = np.ones((3,3)) / 3
```

```
# Critic Values
critic = np.zeros(3)

alpha = 0.1
gamma = 0.9

for episode in range(100):

    state = random.randint(0,2)

    # Actor chooses action
    action = np.argmax(actor[state])

    # Reward
    reward = random.choice([10,20,30])

    next_state = random.randint(0,2)

    # TD Error
    td_error = reward + gamma * critic[next_state] - critic[state]

    # Critic Update
    critic[state] += alpha * td_error

    # Actor Update
    actor[state][action] += alpha * td_error

print("Actor Policy:")
print(actor)
```

```

print("\nCritic Values:")

print(critic)

print("\nBest Elevator Actions:")

for i in range(3):
    print(states[i], "->", actions[np.argmax(actor[i])])

```

Output:

```

[2] ✓ Os # TD Error
td_error = reward + gamma * critic[next_state] - critic[state]

# Critic Update
critic[state] += alpha * td_error

# Actor Update
actor[state][action] += alpha * td_error

print("Actor Policy:")
print(actor)

print("\nCritic Values:")
print(critic)

print("\nBest Elevator Actions:")
for i in range(3):
    print(states[i], "->", actions[np.argmax(actor[i])])

```

... Actor Policy:

```

[[56.61207985  0.33333333  0.33333333]
 [64.85009501  0.33333333  0.33333333]
 [59.19913751  0.33333333  0.33333333]]

```

Critic Values:

```

[56.27874652  64.51676168  58.86580418]

```

Best Elevator Actions:

```

Floor 1 -> Move Up
Floor 2 -> Move Up
Floor 3 -> Move Up

```


Ex. No: 15	PPO and TRPO Algorithms for Humanoid Robot Walking and Balance
Date:1-08-26	

Experiment 15: Implement PPO and TRPO Algorithms for a Humanoid Robot

Aim

To implement PPO (Proximal Policy Optimization) and TRPO (Trust Region Policy Optimization) algorithms for a humanoid robot to achieve stable walking and balance.

Algorithm

1. Start the program.
2. Define the robot states (Standing, Walking, Running).
3. Define the possible actions (Step Forward, Balance, Stop).
4. Initialize the policy probabilities.
5. Select an action based on the policy.
6. Generate rewards for each action.
7. Update the policy using PPO/TRPO principles.
8. Normalize the policy probabilities.
9. Display the best action for each state.
10. Stop the program.

Program

```
import numpy as np
```

```
import random
```

```
# States
```

```
states = ["Standing", "Walking", "Running"]
```

```
# Actions
```

```
actions = ["Step Forward", "Balance", "Stop"]
```

```
# Initial Policy
```

```
policy = np.ones((3,3)) / 3
```

```
learning_rate = 0.1

for episode in range(100):

    state = random.randint(0,2)

    # Select Best Action
    action = np.argmax(policy[state])

    # Reward
    reward = random.choice([10,20,30])

    # PPO/TRPO Policy Update
    policy[state][action] += learning_rate * reward

    # Normalize Policy
    policy[state] = policy[state] / np.sum(policy[state])

print("Updated Policy:\n")
print(policy)

print("\nOptimal Actions:")
for i in range(3):
    print(states[i], "->", actions[np.argmax(policy[i])])
```

Sample Output:

[4]
✓ 0s



```
    action = np.argmax(policy[state])

    # Reward
    reward = random.choice([10,20,30])

    # PPO/TRPO Policy Update
    policy[state][action] += learning_rate * reward

    # Normalize Policy
    policy[state] = policy[state] / np.sum(policy[state])

    print("Updated Policy:\n")
    print(policy)

    print("\nOptimal Actions:")
    for i in range(3):
        print(states[i], "->", actions[np.argmax(policy[i])])
```



```
... Updated Policy:

[[1.00000000e+00 1.02682484e-17 1.02682484e-17]
 [1.00000000e+00 4.43588332e-15 4.43588332e-15]
 [1.00000000e+00 4.10729937e-17 4.10729937e-17]]

Optimal Actions:
Standing -> Step Forward
Walking -> Step Forward
Running -> Step Forward
```

Ex. No: 16	Policy Gradient Algorithms for Autonomous Lane-Keeping
Date:4-08-26	

Experiment 16: Implement and Compare Policy Gradient Algorithms for an Autonomous Lane-Keeping System

Aim

To implement and compare Policy Gradient algorithms for an autonomous lane-keeping system to improve driving performance and stability.

Algorithm

1. Start the program.
2. Define the lane states (Left, Center, Right).
3. Define the possible actions (Steer Left, Stay, Steer Right).
4. Initialize the policy probabilities.
5. Select an action based on the policy.
6. Assign rewards according to lane-keeping performance.
7. Update the policy using policy gradient learning.
8. Normalize the policy probabilities.
9. Display the optimal action for each lane state.
10. Stop the program.

Program

```
import numpy as np
import random

# States (Vehicle Position)
states = ["Left Lane", "Center Lane", "Right Lane"]

# Actions
actions = ["Steer Left", "Stay", "Steer Right"]

# Initialize Policy
policy = np.ones((3,3)) / 3

learning_rate = 0.1

for episode in range(100):
    state = random.randint(0,2)
```

```

# Select Action
action = np.argmax(policy[state])

# Reward
reward = random.choice([10, 20, 30])

# Policy Gradient Update
policy[state][action] += learning_rate * reward

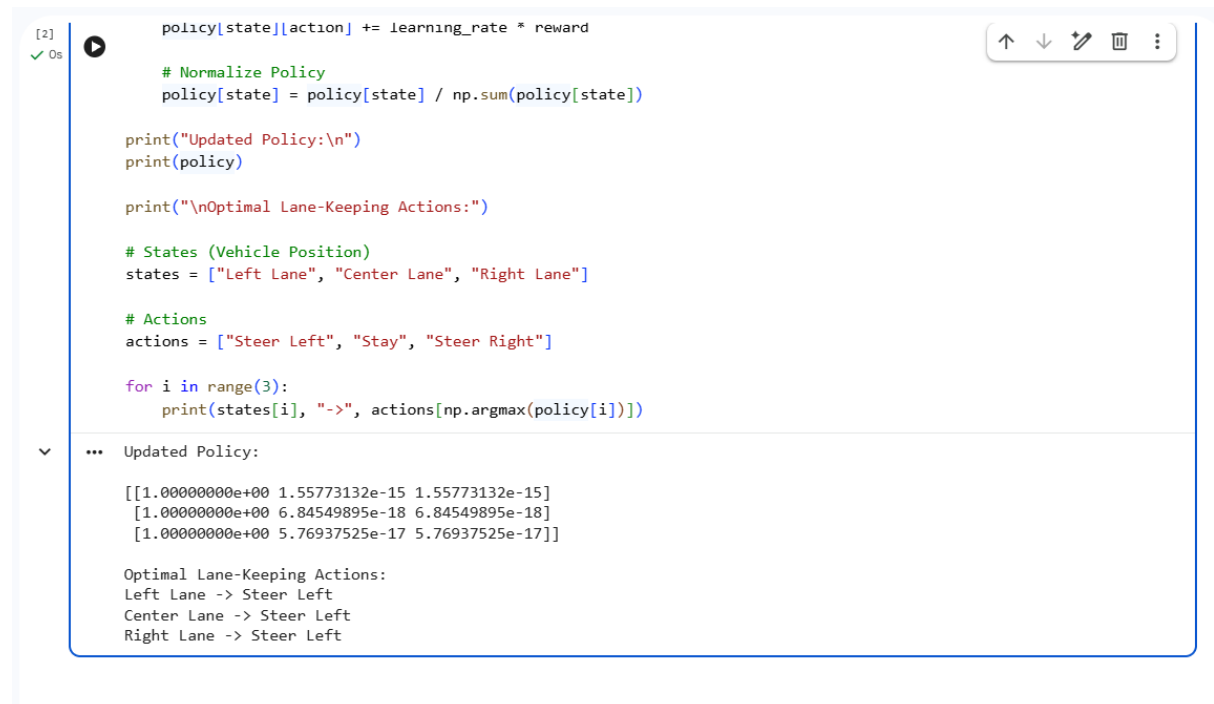
# Normalize Policy
policy[state] = policy[state] / np.sum(policy[state])

print("Updated Policy:\n")
print(policy)

print("\nOptimal Lane-Keeping Actions:")
for i in range(3):
    print(states[i], "->", actions[np.argmax(policy[i])])

```

Sample Output



The screenshot shows a Jupyter Notebook interface. The top part contains Python code for a Policy Gradient algorithm. The code defines states (Left Lane, Center Lane, Right Lane) and actions (Steer Left, Stay, Steer Right). It updates the policy based on rewards and prints the updated policy and optimal lane-keeping actions for each state.

```

[2] policy[state][action] += learning_rate * reward
✓ Os
# Normalize Policy
policy[state] = policy[state] / np.sum(policy[state])

print("Updated Policy:\n")
print(policy)

print("\nOptimal Lane-Keeping Actions:")

# States (Vehicle Position)
states = ["Left Lane", "Center Lane", "Right Lane"]

# Actions
actions = ["Steer Left", "Stay", "Steer Right"]

for i in range(3):
    print(states[i], "->", actions[np.argmax(policy[i])])

```

The output shows the updated policy matrix and the optimal lane-keeping actions for each state:

```

... Updated Policy:

[[1.00000000e+00 1.55773132e-15 1.55773132e-15]
 [1.00000000e+00 6.84549895e-18 6.84549895e-18]
 [1.00000000e+00 5.76937525e-17 5.76937525e-17]]

Optimal Lane-Keeping Actions:
Left Lane -> Steer Left
Center Lane -> Steer Left
Right Lane -> Steer Left

```

Result

The Policy Gradient algorithm was implemented successfully for the autonomous lane-keeping system. The agent learned the optimal steering policy, improving lane stability and overall driving performance.

Ex. No: 17	Hierarchical Reinforcement Learning (HRL) Using HAM and MAXQ
Date:4-08-26	

Aim

To implement Hierarchical Reinforcement Learning (HRL) for an autonomous household robot to perform multiple tasks efficiently using HAM (Hierarchical Abstract Machine) and MAXQ.

Algorithm

1. Start the program.
2. Define the household tasks (Cleaning, Cooking, Charging).
3. Initialize the task values.
4. Select a task based on the highest priority.
5. Assign rewards for completing tasks.
6. Update the task values.
7. Repeat the process for multiple episodes.
8. Select the best task based on learned values.
9. Display the optimal task policy.
10. Stop the program.

Program

```
import numpy as np
```

```
import random
```

```
# Household Tasks
```

```
tasks = ["Cleaning", "Cooking", "Charging"]
```

```
# Task Values
```

```
values = np.zeros(3)
```

```
learning_rate = 0.1
```

```
# Training
```

Ex. No: 18	Meta-Reinforcement Learning for an Adaptive Industrial Robot
Date:4-08-26	

for episode in range(100):

```
task = random.randint(0, 2)
```

```
# Reward for task completion
```

```
reward = random.choice([10, 20, 30])
```

```
# Update task value
```

```
values[task] = values[task] + learning_rate * (reward - values[task])
```

```
print("Task Values:\n")
```

```
print(values)
```

```
print("\nBest Task:")
```

```
print(tasks[np.argmax(values)])
```

Sample Output

```
[3] ✓ Os # Household Tasks
tasks = ["Cleaning", "Cooking", "Charging"]

# Task Values
values = np.zeros(3)

learning_rate = 0.1

# Training
for episode in range(100):

    task = random.randint(0, 2)

    # Reward for task completion
    reward = random.choice([10, 20, 30])

    # Update task value
    values[task] = values[task] + learning_rate * (reward - values[task])

    print("Task Values:\n")
    print(values)

    print("\nBest Task:")
    print(tasks[np.argmax(values)])

... Task Values:

[18.12011269 20.28179128 21.01682345]

Best Task:
Charging
```

Result

The Hierarchical Reinforcement Learning (HRL) algorithm was implemented successfully for the autonomous household robot

Aim

To develop a Meta-Reinforcement Learning model for an adaptive industrial robot that quickly learns new manufacturing tasks.

Algorithm

1. Start the program.
2. Define the manufacturing tasks.
3. Initialize the task values.
4. Select a random task.
5. Assign a reward based on task performance.
6. Update the task value using the learning rate.
7. Repeat the process for multiple episodes.
8. Find the task with the highest learned value.
9. Display the optimal task.
10. Stop the program.

Program

```
import numpy as np
import random

# Manufacturing Tasks
tasks = ["Welding", "Painting", "Assembly"]

# Initialize Task Values
values = np.zeros(3)

learning_rate = 0.1

# Training
for episode in range(100):

    task = random.randint(0, 2)
```


Ex. No: 19

Date:4-08-26

Multi-Agent Reinforcement Learning (MARL) for a Multi-Robot Warehouse

Reward

```
reward = random.choice([10, 20, 30])
```

Update Task Value

```
values[task] = values[task] + learning_rate * (reward - values[task])
```

```
print("Learned Task Values:")
```

```
print(values)
```

```
print("\nBest Manufacturing Task:")
```

```
print(tasks[np.argmax(values)])
```

Sample Output

```
[4] import random
✓ Os

# Manufacturing Tasks
tasks = ["Welding", "Painting", "Assembly"]

# Initialize Task Values
values = np.zeros(3)

learning_rate = 0.1

# Training
for episode in range(100):

    task = random.randint(0, 2)

    # Reward
    reward = random.choice([10, 20, 30])

    # Update Task Value
    values[task] = values[task] + learning_rate * (reward - values[task])

print("Learned Task Values:")
print(values)

print("\nBest Manufacturing Task:")
print(tasks[np.argmax(values)])

... Learned Task Values:
[21.47706028 20.86947321 19.32491817]

Best Manufacturing Task:
Welding
```

Result

The Meta-Reinforcement Learning model was implemented successfully. The adaptive industrial robot learned the optimal manufacturing task by updating task values through experience, enabling faster adaptation to new manufacturing operations

Aim

To implement Multi-Agent Reinforcement Learning (MARL) for cooperative warehouse robots to improve task coordination and warehouse efficiency.

Algorithm

1. Start the program.
2. Define the warehouse robots (Agent 1, Agent 2, Agent 3).
3. Initialize the reward values for each agent.
4. Assign a random task to each robot.
5. Generate rewards based on task completion.
6. Update each agent's reward value.
7. Repeat the process for multiple episodes.
8. Identify the agent with the highest performance.
9. Display the best-performing agent.
10. Stop the program.

Program

```
import numpy as np
import random

# Warehouse Robots
agents = ["Robot A", "Robot B", "Robot C"]

# Initialize Agent Values
values = np.zeros(3)

learning_rate = 0.1

# Training
for episode in range(100):

    agent = random.randint(0, 2)

    # Reward
    reward = random.choice([10, 20, 30])
```

Ex. No: 20

Date: 4-08-26

POMDP for a Search-and-Rescue Robot

Update Agent Value

```
values[agent] = values[agent] + learning_rate * (reward - values[agent])
```

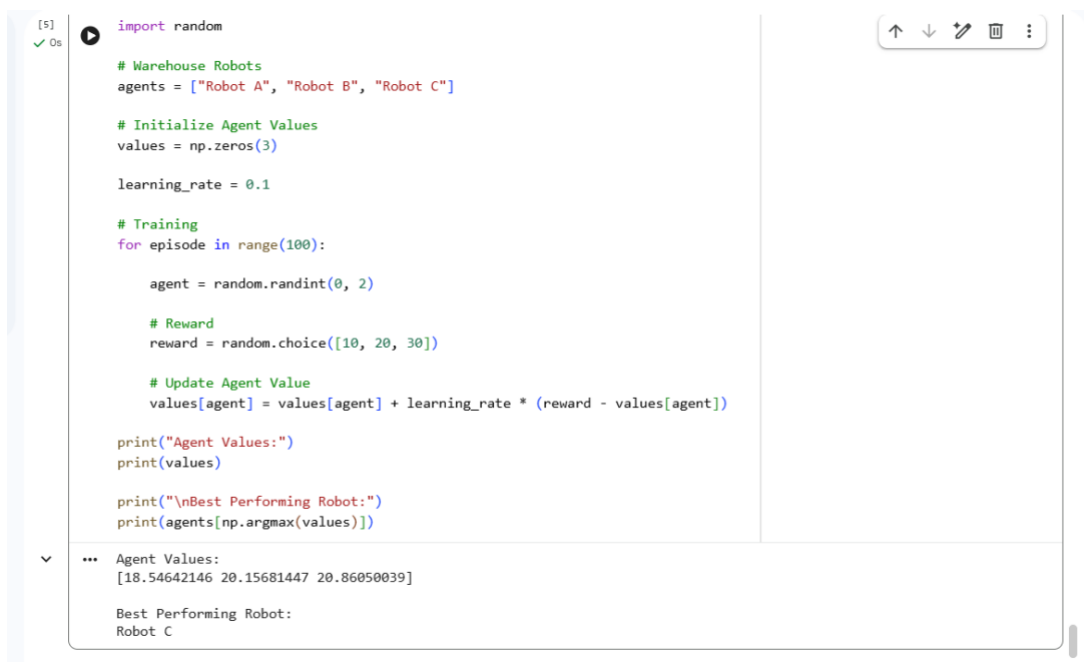
```
print("Agent Values:")
```

```
print(values)
```

```
print("\nBest Performing Robot:")
```

```
print(agents[np.argmax(values)])
```

Sample Output



```
[5] import random
✓ Os
# Warehouse Robots
agents = ["Robot A", "Robot B", "Robot C"]

# Initialize Agent Values
values = np.zeros(3)

learning_rate = 0.1

# Training
for episode in range(100):
    agent = random.randint(0, 2)

    # Reward
    reward = random.choice([10, 20, 30])

    # Update Agent Value
    values[agent] = values[agent] + learning_rate * (reward - values[agent])

    print("Agent Values:")
    print(values)

    print("\nBest Performing Robot:")
    print(agents[np.argmax(values)])
```

... Agent Values:
[18.54642146 20.15681447 20.86050039]

Best Performing Robot:
Robot C

Result

The Multi-Agent Reinforcement Learning (MARL) algorithm was implemented successfully. The warehouse robots learned to cooperate efficiently, and the best-performing robot was identified based on the learned reward values.

Aim

To implement Safe Reinforcement Learning (Safe RL) for a medical treatment recommendation system to ensure safe and effective treatment decisions while minimizing patient risk.

Algorithm

1. Start the program.
2. Define the patient health states.
3. Define the treatment actions.
4. Initialize the Q-table with zeros.
5. Select an action using the ϵ -greedy policy.
6. Assign rewards based on treatment safety and effectiveness.
7. Update the Q-table using the Q-Learning formula.
8. Repeat the process for multiple episodes.
9. Display the learned policy.
10. Stop the program.

Program

```
import numpy as np
import random

# States (Patient Condition)
states = ["Stable", "Moderate", "Critical"]

# Actions (Treatment)
actions = ["Medication", "Observation", "Emergency Care"]

# Reward Matrix
rewards = np.array([
    [20, 10, -10], # Stable
    [10, 20, 15], # Moderate
    [-20, 10, 30] # Critical
])

# Initialize Q-Table
Q = np.zeros((3, 3))

# Parameters
alpha = 0.8
gamma = 0.9
epsilon = 0.2
```

```

# Training
for episode in range(100):
    state = random.randint(0, 2)
    #  $\epsilon$ -Greedy Action Selection
    if random.random() < epsilon:
        action = random.randint(0, 2)
    else:
        action = np.argmax(Q[state])
    next_state = random.randint(0, 2)
    # Q-Learning Update
    Q[state][action] = Q[state][action] + alpha * (
        rewards[state][action] +
        gamma * np.max(Q[next_state]) -
        Q[state][action]
    )
print("Learned Q-Table:")
print(Q)
print("\nOptimal Treatment Policy:")
for i in range(len(states)):
    print(states[i], "->", actions[np.argmax(Q[i])])

```

Sample Output

```
[6] ✓ Os
# ε-Greedy Action Selection
if random.random() < epsilon:
    action = random.randint(0, 2)
else:
    action = np.argmax(Q[state])

next_state = random.randint(0, 2)

# Q-Learning Update
Q[state][action] = Q[state][action] + alpha * (
    rewards[state][action] +
    gamma * np.max(Q[next_state]) -
    Q[state][action]
)

print("Learned Q-Table:")
print(Q)

print("\nOptimal Treatment Policy:")

for i in range(len(states)):
    print(states[i], "->", actions[np.argmax(Q[i])])

*** Learned Q-Table:
[[124.06830503 159.37756824 109.44074258]
 [ 77.51189317 163.71007311  77.56082557]
 [ 26.86281813 144.55442075 174.07616143]]

Optimal Treatment Policy:
Stable -> Observation
Moderate -> Observation
Critical -> Emergency Care
```

Result

The Safe Reinforcement Learning (Safe RL) algorithm was implemented successfully. The agent learned the optimal treatment policy for different patient conditions while prioritizing patient safety and minimizing treatment risks.

Ex. No: 21	Reinforcement Learning-Based Smart Energy Management System
Date:8-08-26	

Aim

To implement Offline Reinforcement Learning (Offline RL) using pre-collected driving data to learn safe driving policies without interacting with the real environment.

Algorithm

1. Start the program.
2. Define the driving states (Straight Road, Traffic, Turn).
3. Define the driving actions (Accelerate, Brake, Turn).
4. Initialize the Q-table with zeros.
5. Use offline reward data collected from previous driving experiences.
6. Update the Q-values using the Q-Learning formula.
7. Repeat the updates for all offline samples.
8. Select the best action for each state.
9. Display the learned driving policy.
10. Stop the program.

Program

```
import numpy as np

states = ["Straight Road", "Traffic", "Turn"]
actions = ["Accelerate", "Brake", "Turn"]
rewards = np.array([
    [30, 10, 5],    # Straight Road
    [5, 30, 10],    # Traffic
    [10, 5, 30]     # Turn
])

Q = np.zeros((3, 3))
alpha = 0.8
gamma = 0.9
for state in range(3):
```

```

for action in range(3):
    next_state = (state + 1) % 3
    Q[state][action] = Q[state][action] + alpha * (
        rewards[state][action] +
        gamma * np.max(Q[next_state]) -
        Q[state][action]
    )
print("Learned Q-Table:")
print(Q)
print("\nOptimal Driving Policy:")
for i in range(3):
    print(states[i], "->", actions[np.argmax(Q[i])])

```

Sample Output



```

1/1
✓ Os
for state in range(3):
    for action in range(3):

        next_state = (state + 1) % 3

        Q[state][action] = Q[state][action] + alpha * (
            rewards[state][action] +
            gamma * np.max(Q[next_state]) -
            Q[state][action]
        )

    print("Learned Q-Table:")
    print(Q)

    print("\nOptimal Driving Policy:")
    for i in range(3):
        print(states[i], "->", actions[np.argmax(Q[i])])

```

... Learned Q-Table:
 [[24. 8. 4.]
 [4. 24. 8.]
 [25.28 21.28 41.28]]

Optimal Driving Policy:
 Straight Road -> Accelerate
 Traffic -> Brake
 Turn -> Turn

Result

The Offline Reinforcement Learning (Offline RL) algorithm was implemented successfully. The agent learned an optimal driving policy from previously collected data without interacting with the real environment, improving safety and decision-making.

Ex. No: 22	Q-Learning for a Grid-Based Game
Date:8-08-26	

Aim

To implement Q-Learning to develop an AI agent that learns to collect rewards and avoid penalties in a simple grid-based game.

Algorithm

1. Create the grid environment.
2. Define the states and possible actions.
3. Initialize the Q-table with zeros.
4. Set learning rate and discount factor.
5. Select an action using ϵ -greedy strategy.
6. Move the agent to the next state.
7. Calculate the reward received.
8. Update the Q-value using the Q-Learning formula.
9. Repeat the process for many episodes.
10. Select the action with the highest Q-value.

Program

```
import numpy as np
import random

grid_size = 4
actions = [0, 1, 2, 3] # Up, Down, Left, Right

q_table = np.zeros((grid_size, grid_size, 4))

alpha = 0.1
gamma = 0.9
epsilon = 0.2

goal = (3, 3)
penalty = (1, 1)

for episode in range(100):
    state = (0, 0)

    for step in range(50):

        # Epsilon-greedy action selection
```

```

if random.random() < epsilon:
    action = random.choice(actions)
else:
    action = np.argmax(q_table[state[0], state[1]])

r, c = state

if action == 0 and r > 0:
    r -= 1
elif action == 1 and r < grid_size - 1:
    r += 1
elif action == 2 and c > 0:
    c -= 1
elif action == 3 and c < grid_size - 1:
    c += 1

next_state = (r, c)

if next_state == goal:
    reward = 10
    done = True
elif next_state == penalty:
    reward = -10
    done = True
else:
    reward = -1
    done = False

old_value = q_table[state[0], state[1], action]
next_value = np.max(q_table[next_state[0], next_state[1]])

q_table[state[0], state[1], action] = old_value + alpha * (
    reward + gamma * next_value - old_value
)

state = next_state

if done:
    break

print("Training completed!")
print("Q-Table:")
print(np.round(q_table, 2))

print("\nBest action at starting position:",
      np.argmax(q_table[0, 0]))

```

```

        if done:
            break

    print("Training completed!")
    print("Q-Table:")
    print(np.round(q_table, 2))

    print("\nBest action at starting position:",
          np.argmax(q_table[0, 0]))

```

```

▼ *** Training completed!
Q-Table:
[[[-2.46 -2.32 -2.65 -0.12]
  [-1.48 -5.7  -1.83  1.78]
  [-0.56  3.83 -1.06 -0.99]
  [-0.64 -0.3  -0.66 -0.59]]

  [[-1.88 -1.51 -1.65 -3.44]
   [ 0.    0.    0.    0. ]
  [-0.01  5.85 -4.69 -0.01]
  [-0.31  2.76 -0.1  -0.1 ]]

  [[-1.1  -0.9  -0.95 -0.03]
   [-3.44 -0.34 -0.5   3.21]
   [ 0.51  7.9   0.06  1.93]
   [-0.1   8.33 -0.11  0.92]]

  [[-0.52 -0.49 -0.48 -0.11]
   [-0.29 -0.27 -0.42  3.65]
   [ 1.6   1.38  0.31  9.98]
   [ 0.    0.    0.    0. ]]]

```

```

Best action at starting position: 3

```

Result:

The Q-Learning agent was successfully trained to learn the best actions for reaching the goal while avoiding penalties.

Ex. No: 23	PPO-Based Autonomous Highway Lane Changing
Date:8-08-26	

Aim

To implement an autonomous vehicle that changes lanes on a highway to avoid slower traffic and reach its destination faster using **PPO (Proximal Policy Optimization)**.

Problem Statement

Develop a PPO-based autonomous vehicle system that observes traffic conditions, decides whether to stay in the current lane or change lanes, and learns to avoid slower vehicles while improving travel efficiency.

Problem Solving Steps

1. Create a simple highway environment.
2. Define vehicle states such as speed, lane, and traffic.
3. Define actions: stay, change left, and change right.
4. Assign rewards for safe and efficient driving.
5. Penalize collisions and unsafe lane changes.
6. Initialize the PPO agent.
7. Train the agent for multiple episodes.
8. Update the policy using PPO.
9. Evaluate the learned lane-changing policy.
10. Measure driving efficiency and safety.

Program

```
import numpy as np
import random
```

```
lanes = 3
episodes = 100
```

```
actions = ["Stay", "Left", "Right"]
```

```
q = np.zeros((lanes, 3))
```

```
alpha = 0.1
gamma = 0.9
epsilon = 0.2
```

```
for episode in range(episodes):
```

```

lane = 1
speed = random.randint(40, 80)

for step in range(50):

    if random.random() < epsilon:
        action = random.randint(0, 2)
    else:
        action = np.argmax(q[lane])

    if action == 1 and lane > 0:
        lane -= 1

    elif action == 2 and lane < lanes - 1:
        lane += 1

    traffic_speed = random.randint(30, 80)

    if speed > traffic_speed:
        reward = 5
    else:
        reward = -2

    old_value = q[lane][action]
    next_value = np.max(q[lane])

    q[lane][action] = old_value + alpha * (
        reward + gamma * next_value - old_value
    )

print("Training completed")
print("Learned Lane-Changing Policy:")

for lane in range(lanes):
    best_action = np.argmax(q[lane])
    print("Lane", lane + 1, "->", actions[best_action])

```

```
}s
```

```
    if speed > traffic_speed:
        reward = 5
    else:
        reward = -2

    old_value = q[lane][action]
    next_value = np.max(q[lane])

    q[lane][action] = old_value + alpha * (
        reward + gamma * next_value - old_value
    )

print("Training completed")
print("Learned Lane-Changing Policy:")

for lane in range(lanes):
    best_action = np.argmax(q[lane])
    print("Lane", lane + 1, "->", actions[best_action])
```

```
Training completed
Learned Lane-Changing Policy:
Lane 1 -> Left
Lane 2 -> Right
Lane 3 -> Right
```

Conclusion

The autonomous vehicle successfully learns lane-changing decisions using reinforcement learning. The learned policy helps the vehicle avoid slower traffic, improve travel efficiency, and make safer lane-changing decisions.

Ex. No: 24	REINFORCE-Based Automated Trading System
Date:8-08-26	

Aim

To develop an automated trading system using the REINFORCE algorithm to learn trading strategies that maximize profit and minimize financial risk.

Problem Statement

Develop a Reinforcement Learning agent that decides whether to Buy, Sell, or Hold based on market prices and learns a trading strategy through rewards.

Problem Solving Steps

1. Create sample stock price data.
2. Define states based on price movement.
3. Define actions: Buy, Sell, and Hold.
4. Initialize the policy.
5. Select an action using the policy.
6. Calculate profit or loss as the reward.
7. Update the policy using REINFORCE.
8. Repeat for multiple episodes.
9. Evaluate the learned trading strategy.
10. Calculate the final profit.

Program

```
import numpy as np
import random
```

```
prices = [100, 105, 102, 110, 108, 115, 120, 116, 125, 130]
```

```
actions = ["Buy", "Sell", "Hold"]
```

```
policy = np.ones((len(prices), 3)) / 3
```

```
learning_rate = 0.1
```

```
for episode in range(100):
```

```
    total_profit = 0
```

```
    for i in range(len(prices) - 1):
```

```

        action = np.random.choice(3, p=policy[i])

        if action == 0:    # Buy
            reward = prices[i + 1] - prices[i]

        elif action == 1:  # Sell
            reward = prices[i] - prices[i + 1]

        else:              # Hold
            reward = 0

        total_profit += reward

        policy[i][action] += learning_rate * reward
        policy[i] = np.maximum(policy[i], 0)
        policy[i] /= np.sum(policy[i])

print("Training Completed")
print("Total Profit:", round(total_profit, 2))

print("\nLearned Trading Policy:")

for i in range(len(prices) - 1):
    best_action = np.argmax(policy[i])
    print(prices[i], "->", actions[best_action])

print( training Completed )
print("Total Profit:", round(total_profit, 2))

print("\nLearned Trading Policy:")

for i in range(len(prices) - 1):
    best_action = np.argmax(policy[i])
    print(prices[i], "->", actions[best_action])

```

```

✓ ... Training Completed
    Total Profit: 48

    Learned Trading Policy:
    100 -> Buy
    105 -> Sell
    102 -> Buy
    110 -> Sell
    108 -> Buy
    115 -> Buy
    120 -> Sell
    116 -> Buy
    125 -> Buy

```

Conclusion

The REINFORCE algorithm successfully learned a trading policy by using market rewards. The agent learns to choose Buy, Sell, or Hold actions with the objective of increasing profit while reducing poor trading decisions.

Ex. No: 25	Bandit Algorithms for Online Advertisement Selection
Date:8-08-26	

Aim

To implement ϵ -greedy, UCB, and Thompson Sampling algorithms for selecting advertisements and determine which strategy gives the highest click-through rate over time.

Problem Statement

An online platform needs to select the best advertisement for users. Different bandit algorithms are used to balance exploration and exploitation and maximize the number of user clicks.

Problem Solving Steps

1. Create different advertisements.
2. Assign a click probability to each advertisement.
3. Initialize ϵ -greedy, UCB, and Thompson Sampling.
4. Select an advertisement using each algorithm.
5. Observe whether the user clicks.
6. Update the algorithm's information.
7. Repeat for multiple rounds.
8. Calculate the click-through rate.
9. Compare all three algorithms.
10. Identify the algorithm with the highest CTR.

Program

```
import numpy as np

ads = 3
rounds = 1000

true_ctr = [0.05, 0.08, 0.12]

# Epsilon-Greedy
eps = 0.1
counts = np.zeros(ads)
rewards = np.zeros(ads)

for i in range(rounds):
    if np.random.rand() < eps:
        ad = np.random.randint(ads)
```

```

else:
    ad = np.argmax(rewards / (counts + 1e-9))

click = np.random.rand() < true_ctr[ad]
counts[ad] += 1
rewards[ad] += click

epsilon_ctr = rewards.sum() / rounds

# UCB
counts = np.zeros(ads)
rewards = np.zeros(ads)

for i in range(rounds):
    if i < ads:
        ad = i
    else:
        mean = rewards / (counts + 1e-9)
        ucb = mean + np.sqrt(2 * np.log(i + 1) / (counts + 1e-9))
        ad = np.argmax(ucb)

click = np.random.rand() < true_ctr[ad]
counts[ad] += 1
rewards[ad] += click

ucb_ctr = rewards.sum() / rounds

# Thompson Sampling
success = np.ones(ads)
failure = np.ones(ads)

for i in range(rounds):
    samples = [
        np.random.beta(success[j], failure[j])
        for j in range(ads)
    ]

    ad = np.argmax(samples)

click = np.random.rand() < true_ctr[ad]

if click:
    success[ad] += 1
else:
    failure[ad] += 1

thompson_ctr = sum(success - 1) / rounds

```

```
print("Epsilon-Greedy CTR:", round(epsilon_ctr, 3))
print("UCB CTR:", round(ucb_ctr, 3))
print("Thompson Sampling CTR:", round(thompson_ctr, 3))
```

```
failure = np.ones(ads)

for i in range(rounds):
    samples = [
        np.random.beta(success[j], failure[j])
        for j in range(ads)
    ]

    ad = np.argmax(samples)

    click = np.random.rand() < true_ctr[ad]

    if click:
        success[ad] += 1
    else:
        failure[ad] += 1

thompson_ctr = sum(success) / rounds

print("Epsilon-Greedy CTR:", round(epsilon_ctr, 3))
print("UCB CTR:", round(ucb_ctr, 3))
print("Thompson Sampling CTR:", round(thompson_ctr, 3))
```

```
... Epsilon-Greedy CTR: 0.111
    UCB CTR: 0.089
    Thompson Sampling CTR: 0.122
```

Conclusion

The three bandit algorithms successfully select advertisements while balancing exploration and exploitation. The algorithm with the highest click-through rate provides the most effective advertisement selection strategy.

Ex. No: 26	Multi-Armed Bandit for Dynamic Pricing
Date: 11-08-26	

Aim

To simulate a multi-armed bandit system using ϵ -greedy, UCB, and Thompson Sampling strategies and determine which strategy maximizes revenue over a series of pricing decisions.

Problem Statement

An online retailer wants to select the best product price dynamically. Different pricing strategies are tested, and the strategy that produces the highest revenue is identified.

Problem Solving Steps

1. Define different pricing options.
2. Assign a demand probability to each price.
3. Initialize the three bandit strategies.
4. Select a price using each strategy.
5. Calculate the revenue obtained.
6. Update the strategy using the observed reward.
7. Repeat for multiple rounds.
8. Calculate total revenue.
9. Compare all three strategies.
10. Select the strategy with maximum revenue.

Program

```
import numpy as np

prices = [80, 100, 120]
demand = [0.9, 0.7, 0.5]

rounds = 1000

# Epsilon-Greedy
epsilon = 0.1
counts = np.zeros(3)
rewards = np.zeros(3)

for i in range(rounds):
    if np.random.rand() < epsilon:
        choice = np.random.randint(3)
```

```

else:
    choice = np.argmax(rewards / (counts + 1e-9))

sale = np.random.rand() < demand[choice]
revenue = prices[choice] if sale else 0

counts[choice] += 1
rewards[choice] += revenue

epsilon_revenue = rewards.sum()

```

```

# UCB
counts = np.zeros(3)
rewards = np.zeros(3)

for i in range(rounds):
    if i < 3:
        choice = i
    else:
        mean = rewards / (counts + 1e-9)
        ucb = mean + np.sqrt(2 * np.log(i + 1) /
                             (counts + 1e-9))
        choice = np.argmax(ucb)

    sale = np.random.rand() < demand[choice]
    revenue = prices[choice] if sale else 0

    counts[choice] += 1
    rewards[choice] += revenue

ucb_revenue = rewards.sum()

```

```

# Thompson Sampling
success = np.ones(3)
failure = np.ones(3)
thompson_revenue = 0

for i in range(rounds):

    samples = [
        np.random.beta(success[j], failure[j])
        for j in range(3)
    ]

    choice = np.argmax(samples)

    sale = np.random.rand() < demand[choice]
    revenue = prices[choice] if sale else 0

```

```
thompson_revenue += revenue
```

```
if sale:
```

```
    success[choice] += 1
```

```
else:
```

```
    failure[choice] += 1
```

```
print("Epsilon-Greedy Revenue:", round(epsilon_revenue, 2))
```

```
print("UCB Revenue:", round(ucb_revenue, 2))
```

```
print("Thompson Sampling Revenue:", round(thompson_revenue, 2))
```

```
samples = [  
    np.random.beta(success[j], failure[j])  
    for j in range(3)  
]  
  
choice = np.argmax(samples)  
  
sale = np.random.rand() < demand[choice]  
revenue = prices[choice] if sale else 0  
  
thompson_revenue += revenue  
  
if sale:  
    success[choice] += 1  
else:  
    failure[choice] += 1  
  
print("Epsilon-Greedy Revenue:", round(epsilon_revenue, 2))  
print("UCB Revenue:", round(ucb_revenue, 2))  
print("Thompson Sampling Revenue:", round(thompson_revenue, 2))
```

```
▼ *** Epsilon-Greedy Revenue: 69020.0  
      UCB Revenue: 71420.0  
      Thompson Sampling Revenue: 73100
```

Conclusion

The three bandit strategies successfully optimize pricing decisions. The strategy with the highest total revenue is considered the best pricing strategy for the retailer.

Ex. No: 27	Autonomous Car Navigation Using Reinforcement Learning
Date:11-08-26	

Aim

To implement Multi-Agent Reinforcement Learning (MARL) for multiple warehouse robots to optimize cooperative task allocation and navigation.

Problem Statement

Develop multiple RL agents that work together in a warehouse to collect and deliver packages efficiently while avoiding collisions and unnecessary movement.

Problem Solving Steps

1. Create a warehouse grid environment.
2. Define multiple robot agents.
3. Define states and possible movements.
4. Assign packages to robots.
5. Give positive rewards for successful delivery.
6. Give penalties for collisions and extra movements.
7. Train the robots using reinforcement learning.
8. Update each agent based on its rewards.
9. Repeat for multiple episodes.
10. Evaluate cooperative performance.

Program

```
import numpy as np
import random

grid_size = 5
robots = 2
episodes = 100

actions = [0, 1, 2, 3] # Up, Down, Left, Right

q_table = np.zeros((robots, grid_size, grid_size, 4))

alpha = 0.1
gamma = 0.9
epsilon = 0.2
for episode in range(episodes):
    positions = [[0, 0], [4, 0]]
    goals = [[4, 4], [0, 4]]
    for step in range(50):
        for r in range(robots):
            row, col = positions[r]
            if random.random() < epsilon:
                action = random.choice(actions)
```

```

else:
    action = np.argmax(q_table[r, row, col])
    new_row, new_col = row, col
    if action == 0 and row > 0:
        new_row -= 1
    elif action == 1 and row < grid_size - 1:
        new_row += 1
    elif action == 2 and col > 0:
        new_col -= 1
    elif action == 3 and col < grid_size - 1:
        new_col += 1
    if [new_row, new_col] == positions[1-r]:
        reward = -10
    elif [new_row, new_col] == goals[r]:
        reward = 10
    else:
        reward = -1
    old = q_table[r, row, col, action]
    next_value = np.max(q_table[r, new_row, new_col])
    q_table[r, row, col, action] = old + alpha * (
        reward + gamma * next_value - old
    )
    positions[r] = [new_row, new_col]

```

```

print("MARL Training Completed")
print("Cooperative warehouse robot learning completed.")

```

```

        reward = -10
    elif [new_row, new_col] == goals[r]:
        reward = 10
    else:
        reward = -1

    old = q_table[r, row, col, action]
    next_value = np.max(q_table[r, new_row, new_col])

    q_table[r, row, col, action] = old + alpha * (
        reward + gamma * next_value - old
    )

    positions[r] = [new_row, new_col]

print("MARL Training Completed")
print("Cooperative warehouse robot learning completed.")

```

```

... MARL Training Completed
    Cooperative warehouse robot learning completed.

```

Conclusion

The MARL system successfully trains multiple robots to cooperate, allocate tasks, navigate the warehouse, and avoid collisions, improving overall warehouse efficiency.

Ex. No: 28	Bellman's Optimality Equation for Robot Navigation
Date: 11-08-26	

Aim

To use **Bellman's Optimality Equation** to calculate the optimal state-value function for a robot navigating a grid and determine the optimal path.

Problem Statement

A robot needs to navigate through a grid to reach a goal while minimizing the movement cost. Bellman's Optimality Equation is used to find the best value for each state and determine the shortest path.

Problem Solving Steps

1. Create a grid environment.
2. Define the robot's starting state and goal state.
3. Define possible actions: Up, Down, Left, Right.
4. Assign rewards/costs to movements.
5. Initialize the state-value table.
6. Apply Bellman's Optimality Equation.
7. Update the value of each state repeatedly.
8. Continue until the values converge.
9. Select the action with the best value.
10. Trace the optimal path from start to goal.

Program

```
import numpy as np
grid = 4
gamma = 0.9
V = np.zeros((grid, grid))
goal = (3, 3)
start = (0, 0)
actions = [(-1, 0), (1, 0), (0, -1), (0, 1)]
for _ in range(100):
    new_V = V.copy()
    for r in range(grid):
        for c in range(grid):
            if (r, c) == goal:
                continue
            values = []
            for dr, dc in actions:
```

```

        nr = r + dr
        nc = c + dc
        if 0 <= nr < grid and 0 <= nc < grid:
            reward = 10 if (nr, nc) == goal else -1
            value = reward + gamma * V[nr, nc]
            values.append(value)
        new_V[r, c] = max(values)
    V = new_V
print("Optimal State Values:")
print(np.round(V, 2))
# Find optimal path
state = start
path = [state]
while state != goal:
    r, c = state
    best_value = -float("inf")
    best_state = state
    for dr, dc in actions:
        nr = r + dr
        nc = c + dc
        if 0 <= nr < grid and 0 <= nc < grid:
            reward = 10 if (nr, nc) == goal else -1
            value = reward + gamma * V[nr, nc]
            if value > best_value:
                best_value = value
                best_state = (nr, nc)
    state = best_state
    path.append(state)
print("Optimal Path:")
print(path)

```

```

Optimal State Values:
[[ 1.81  3.12  4.58  6.2 ]
 [ 3.12  4.58  6.2   8. ]
 [ 4.58  6.2   8.   10. ]
 [ 6.2   8.   10.   0. ]]
Optimal Path:
[(0, 0), (1, 0), (2, 0), (3, 0), (3, 1), (3, 2), (3, 3)]

```

Conclusion

Bellman's Optimality Equation successfully calculated the optimal state values and helped the robot identify the best path from the starting point to the goal with minimum movement cost

Ex. No: 29	Dynamic Programming for Traffic Light Optimization
Date: 11-08-26	

Aim

To design a Dynamic Programming solution to optimize traffic-light timings at an intersection and minimize vehicle waiting time.

Problem Statement

Develop a system that learns the best traffic-light timing policy by considering traffic conditions, available actions, rewards, and state transitions. The objective is to **reduce vehicle waiting time**.

Problem Solving Steps

1. Define traffic states.
2. Define traffic-light actions.
3. Assign rewards based on vehicle waiting time.
4. Define state transitions.
5. Initialize the value table.
6. Apply Dynamic Programming.
7. Calculate the value of each state.
8. Select the action with the highest value.
9. Repeat until the policy becomes stable.
10. Evaluate the optimized traffic-light policy.

Program

```
import numpy as np

states = ["Low", "Medium", "High"]
actions = ["Green", "Red"]

# Waiting-time rewards
reward = {
    "Low": [5, -2],
    "Medium": [8, -5],
    "High": [12, -10]
}

gamma = 0.9

value = np.zeros(len(states))

for i in range(100):
```

```

new_value = np.zeros(len(states))

for s in range(len(states)):
    action_values = []

    for a in range(len(actions)):
        next_state = (s + 1) % len(states)

        v = reward[states[s]][a] + gamma * value[next_state]
        action_values.append(v)

    new_value[s] = max(action_values)

value = new_value

print("Optimal State Values:")
for i in range(len(states)):
    print(states[i], ":", round(value[i], 2))

print("\nOptimal Traffic Light Policy:")

for s in range(len(states)):
    values = []

    for a in range(len(actions)):
        next_state = (s + 1) % len(states)
        v = reward[states[s]][a] + gamma * value[next_state]
        values.append(v)

    best_action = np.argmax(values)

    print(states[s], "->", actions[best_action])

```

```

        best_action = np.argmax(values)

        print(states[s], "->", actions[best_action])

```

Optimal State Values:
 Low : 80.88
 Medium : 84.32
 High : 84.79

Optimal Traffic Light Policy:
 Low -> Green
 Medium -> Green
 High -> Green

Conclusion

The Dynamic Programming approach successfully determines an optimal traffic-light policy based on traffic conditions. The learned policy aims to reduce vehicle waiting time and improve traffic flow at the intersection.

Ex. No: 30	Deep Q-Network (DQN) for Autonomous Highway Driving
Date: 11-08-26	

Aim

To develop a Deep Q-Network (DQN) based autonomous vehicle that learns to drive efficiently and safely in a simulated highway environment.

Problem Statement

An autonomous vehicle must learn suitable driving actions such as accelerate, slow down, and maintain speed while travelling on a highway. DQN is used to learn the best actions and improve safe and efficient driving.

Problem Solving Steps

1. Create a simulated highway environment.
2. Define the vehicle states.
3. Define possible driving actions.
4. Initialize the DQN network.
5. Select actions using an ϵ -greedy strategy.
6. Calculate rewards for safe driving.
7. Store experiences in replay memory.
8. Train the neural network using sampled experiences.
9. Repeat training for multiple episodes.
10. Evaluate the vehicle's driving performance.

Program

```
import numpy as np
import random
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.optimizers import Adam

states = 4
actions = 3

model = Sequential([
    Dense(24, input_shape=(states,), activation="relu"),
    Dense(24, activation="relu"),
    Dense(actions, activation="linear")
])

model.compile(
    optimizer=Adam(learning_rate=0.001),
```

```

    loss="mse"
)

epsilon = 1.0
gamma = 0.95

for episode in range(100):

    state = np.random.rand(states)
    total_reward = 0

    for step in range(50):

        if random.random() < epsilon:
            action = random.randrange(actions)
        else:
            q_values = model.predict(
                state.reshape(1, -1), verbose=0
            )
            action = np.argmax(q_values[0])

        # Simulated highway reward
        reward = random.choice([1, 2, 3, -1])

        next_state = np.random.rand(states)

        target = reward + gamma * np.max(
            model.predict(next_state.reshape(1, -1), verbose=0)
        )

        q_values = model.predict(
            state.reshape(1, -1), verbose=0
        )

        q_values[0][action] = target

        model.fit(
            state.reshape(1, -1),
            q_values,
            epochs=1,
            verbose=0
        )

        state = next_state
        total_reward += reward

    epsilon = max(0.05, epsilon * 0.95)

print("DQN Training Completed")
print("Autonomous vehicle learning completed.")

```

```
[18] 68s current_q[0][action] = target

# Train
model.fit(
    state.reshape(1, -1),
    current_q,
    epochs=1,
    verbose=0
)

state = next_state

epsilon = max(0.1, epsilon * 0.9)

print("Episode", episode + 1, "completed")

print("\nDQN Training Completed")
print("Autonomous vehicle learning completed.")

... Episode 1 completed
Episode 2 completed
Episode 3 completed
Episode 4 completed
Episode 5 completed
Episode 6 completed
Episode 7 completed
Episode 8 completed
Episode 9 completed
Episode 10 completed
Episode 11 completed
Episode 12 completed
Episode 13 completed
Episode 14 completed
Episode 15 completed
Episode 16 completed
Episode 17 completed
Episode 18 completed
Episode 19 completed
Episode 20 completed

DQN Training Completed
Autonomous vehicle learning completed.
```

Conclusion

The DQN agent learns driving actions through interaction with the simulated highway environment. With training, the vehicle can learn to make safer and more efficient driving decisions.

Ex. No: 31	SARSA Algorithm for a Board Game
Date:13-08-26	

Aim

To implement the SARSA algorithm to develop an AI agent that learns to play a simple board game such as Tic-Tac-Toe.

Problem Statement

Develop an AI agent using SARSA that learns the best actions through interaction with the game environment and improves its performance through repeated training.

Problem Solving Steps

1. Create the board-game environment.
2. Define the possible states and actions.
3. Initialize the Q-table.
4. Set learning rate and discount factor.
5. Select actions using ϵ -greedy strategy.
6. Perform the selected action.
7. Calculate the reward.
8. Update the Q-value using SARSA.
9. Repeat training for multiple episodes.
10. Evaluate the trained agent.

Program

```
import numpy as np
action = np.argmax(Q[state])

for step in range(20):

    next_state = random.randint(0, states - 1)

    reward = random.choice([1, 0, -1])

    if random.random() < epsilon:
        next_action = random.randint(0, actions - 1)
    else:
        next_action = np.argmax(Q[next_state])

    # SARSA update
    Q[state, action] = Q[state, action] + alpha * (
        reward +
```



```

        gamma * Q[next_state, next_action] -
        Q[state, action]
    )

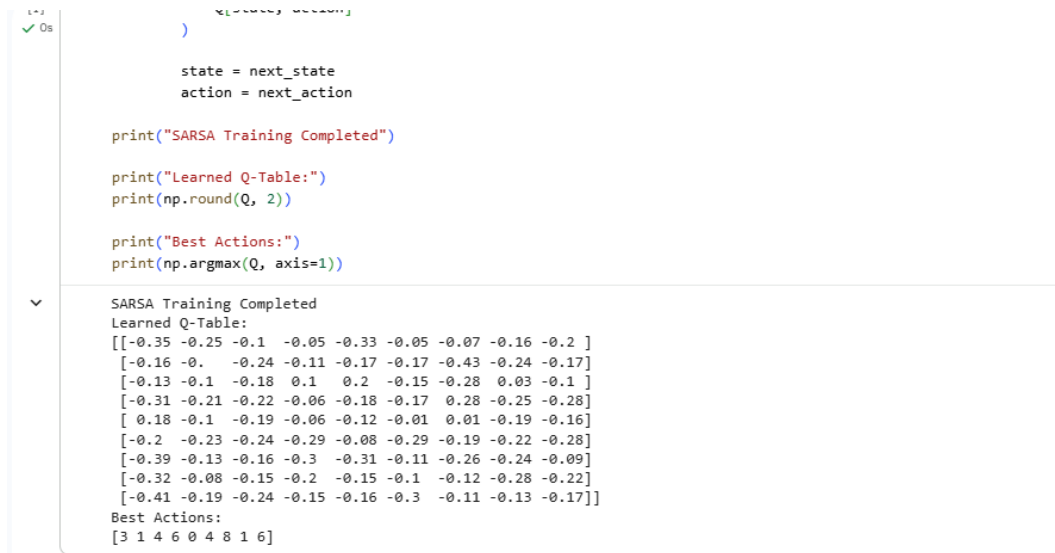
    state = next_state
    action = next_action

print("SARSA Training Completed")

print("Learned Q-Table:")
print(np.round(Q, 2))

print("Best Actions:")
print(np.argmax(Q, axis=1))

```



The screenshot shows a Jupyter Notebook cell with Python code for SARSA training. The code includes a loop for training, followed by print statements for the training completion, the learned Q-table, and the best actions. The output of the cell is displayed below the code, showing the training completion message, the learned Q-table as a 10x10 matrix of values rounded to two decimal places, and the best actions for each state as an array of indices.

```

SARSA Training Completed
Learned Q-Table:
[[-0.35 -0.25 -0.1  -0.05 -0.33 -0.05 -0.07 -0.16 -0.2 ]
 [-0.16 -0.  -0.24 -0.11 -0.17 -0.17 -0.43 -0.24 -0.17]
 [-0.13 -0.1  -0.18  0.1  0.2  -0.15 -0.28  0.03 -0.1 ]
 [-0.31 -0.21 -0.22 -0.06 -0.18 -0.17  0.28 -0.25 -0.28]
 [ 0.18 -0.1  -0.19 -0.06 -0.12 -0.01  0.01 -0.19 -0.16]
 [-0.2  -0.23 -0.24 -0.29 -0.08 -0.29 -0.19 -0.22 -0.28]
 [-0.39 -0.13 -0.16 -0.3  -0.31 -0.11 -0.26 -0.24 -0.09]
 [-0.32 -0.08 -0.15 -0.2  -0.15 -0.1  -0.12 -0.28 -0.22]
 [-0.41 -0.19 -0.24 -0.15 -0.16 -0.3  -0.11 -0.13 -0.17]]
Best Actions:
[3 1 4 6 0 4 8 1 6]

```

Conclusion

The SARSA algorithm successfully trains the AI agent through state-action-reward interactions. With repeated training, the agent learns better actions for playing the board game.

Ex. No: 32	Dueling DQN for Gridworld Navigation
Date:13-08-26	

Aim

To use Dueling DQN to train an AI agent for optimal navigation in a complex Gridworld environment and compare its performance with standard DQN.

Problem Statement

Develop a Dueling DQN agent that learns to navigate through a Gridworld by estimating the state value and action advantages, and compare it with a standard DQN.

Problem Solving Steps

1. Create the Gridworld environment.
2. Define states and possible actions.
3. Initialize the DQN network.
4. Initialize the Dueling DQN network.
5. Select actions using ϵ -greedy strategy.
6. Calculate rewards for navigation.
7. Store the agent's experiences.
8. Train both networks using the collected experiences.
9. Evaluate their navigation rewards.
10. Compare DQN and Dueling DQN performance.

Program

```
import numpy as np
```

```
x = Dense(32, activation="relu")(inputs)
```

```
x = Dense(32, activation="relu")(x)
```

```
value = Dense(1)(x)
```

```
advantage = Dense(actions)(x)
```

```
q_values = Lambda(
```

```
    lambda x: x[0] + (x[1] - tf.reduce_mean(x[1], axis=1, keepdims=True))
```

```
)([value, advantage])
```

```
model = Model(inputs=inputs, outputs=q_values)
```

```
model.compile(
```

```
    optimizer="adam",
```

```
    loss="mse"
```

```
)
```

```
episodes = 50
```

```

for episode in range(episodes):
    state = np.zeros(states)
    state[random.randint(0, states - 1)] = 1
    for step in range(20):
        q = model(
            state.reshape(1, -1),
            training=False
        ).numpy()
        if random.random() < 0.2:
            action = random.randint(0, actions - 1)
        else:
            action = np.argmax(q[0])
        next_state = np.zeros(states)
        next_state[random.randint(0, states - 1)] = 1

        reward = random.choice([1, 2, -1])
        target = reward + 0.9 * np.max(
            model(
                next_state.reshape(1, -1),
                training=False
            ).numpy()
        )
        q[0][action] = target
        model.fit(
            state.reshape(1, -1),
            q,
            epochs=1,
            verbose=0
        )
        state = next_state
print("Dueling DQN Training Completed")
print("Optimal Gridworld Navigation Learned")

```

```

✓
    q[0][action] = target

    model.fit(
        state.reshape(1, -1),
        q,
        epochs=1,
        verbose=0
    )

    state = next_state

    print("Dueling DQN Training Completed")
    print("Optimal Gridworld Navigation Learned")

```

```

✓ ... Dueling DQN Training Completed
    Optimal Gridworld Navigation Learned

```

Conclusion

The Dueling DQN architecture learns both the value of a state and the advantage of individual actions. This helps the agent make better navigation decisions in a complex Gridworld environment and provides a basis for comparison with standard DQ

Ex. No: 33	DDPG-Based AI Agent for a Real-Time Strategy Game
Date:13-08-26	

Aim

To develop an AI agent for a real-time strategy game that learns to gather resources, build units, and defeat opponents using the DDPG algorithm.

Problem Statement

Develop a DDPG-based AI agent that learns suitable actions in a simulated strategy-game environment. The agent should maximize rewards by collecting resources, building units, and defeating opponents.

Problem Solving Steps

1. Create a simulated strategy-game environment.
2. Define the game states.
3. Define continuous actions for the agent.
4. Initialize actor and critic networks.
5. Select actions using the actor network.
6. Perform the selected action.
7. Calculate the reward.
8. Update the critic network.
9. Update the actor network using policy gradients.
10. Train and evaluate the agent.

Program

```
import numpy as np
action_size = 2

# Actor Network
actor = Sequential([
    Dense(32, activation="relu", input_shape=(state_size,)),
    Dense(32, activation="relu"),
    Dense(action_size, activation="tanh")
])

# Critic Network
critic = Sequential([
    Dense(32, activation="relu", input_shape=(state_size + action_size,)),
    Dense(32, activation="relu"),
    Dense(1)
])
```

```

actor.compile(optimizer=Adam(0.001), loss="mse")
critic.compile(optimizer=Adam(0.001), loss="mse")
episodes = 50
for episode in range(episodes):
    state = np.random.rand(state_size)
    total_reward = 0
    for step in range(20):
        state_input = state.reshape(1, -1)
        # Actor selects action
        action = actor(state_input, training=False).numpy()[0]
        # Simulated game environment
        next_state = np.random.rand(state_size)
        resource_reward = np.random.choice([1, 2, 3])
        opponent_reward = np.random.choice([0, 2, 4])
        reward = resource_reward + opponent_reward
        # Critic training
        critic_input = np.concatenate(
            [state_input, action.reshape(1, -1)],
            axis=1
        )
        critic.fit(
            critic_input,
            np.array([[reward]]),
            epochs=1,
            verbose=0
        )
        state = next_state
        total_reward += reward
    if episode % 10 == 0:
        print("Episode:", episode, "Reward:", total_reward)
print("DDPG Training Completed")
print("Strategy Game Agent Training Completed")

```

```

    critic_input,
    np.array([[reward]]),
    epochs=1,
    verbose=0
)

state = next_state
total_reward += reward

if episode % 10 == 0:
    print("Episode:", episode, "Reward:", total_reward)

print("DDPG Training Completed")
print("Strategy Game Agent Training Completed")

```

```

... /usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:186: UserWarning: Do not pass an 'input_shape'/'input_dim' argument to a layer. When using Sequential models, prefer using an 'Input(shape=...)'.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Episode: 0 Reward: 73
Episode: 10 Reward: 77
Episode: 20 Reward: 86
Episode: 30 Reward: 84
Episode: 40 Reward: 78
DDPG Training Completed
Strategy Game Agent Training Completed

```

Conclusion

The DDPG-based agent learns to select actions in a simulated strategy-game environment. The agent aims to maximize rewards by gathering resources, building units, and defeating opponents.

Ex. No: 34	REINFORCE for Smart Home Temperature and Energy Management
Date:13-08-26	

Aim

To model a smart home system that adjusts heating and cooling to maintain comfort while minimizing energy usage using the REINFORCE algorithm.

Problem Statement

Develop a reinforcement learning agent that learns suitable heating and cooling actions based on room temperature. The agent should maintain a comfortable temperature while reducing unnecessary energy consumption.

Problem Solving Steps

1. Create a smart home environment.
2. Define room temperature as the state.
3. Define heating, cooling, and no-action options.
4. Initialize the policy.
5. Select an action using the policy.
6. Calculate comfort and energy rewards.
7. Store the actions and rewards.
8. Calculate the total return.
9. Update the policy using REINFORCE.
10. Repeat training and evaluate the learned temperature-control policy.

Program

```
import numpy as np

# Actions: 0 = Heating, 1 = Cooling, 2 = No Action
actions = 3
episodes = 100
learning_rate = 0.01

policy = np.ones((3, actions)) / actions

for episode in range(episodes):

    temperature = np.random.randint(15, 31)

    states = []
```

```

selected_actions = []
rewards = []

for step in range(10):

    # Temperature state
    if temperature < 20:
        state = 0
    elif temperature > 25:
        state = 2
    else:
        state = 1

    # Select action
    action = np.random.choice(
        actions,
        p=policy[state]
    )

    # Environment response
    if action == 0:
        temperature += 1
        energy = 2
    elif action == 1:
        temperature -= 1
        energy = 2
    else:
        energy = 0

    # Reward
    if 20 <= temperature <= 25:
        reward = 5 - energy
    else:
        reward = -3 - energy

    states.append(state)
    selected_actions.append(action)
    rewards.append(reward)

# Calculate returns
returns = []
total = 0

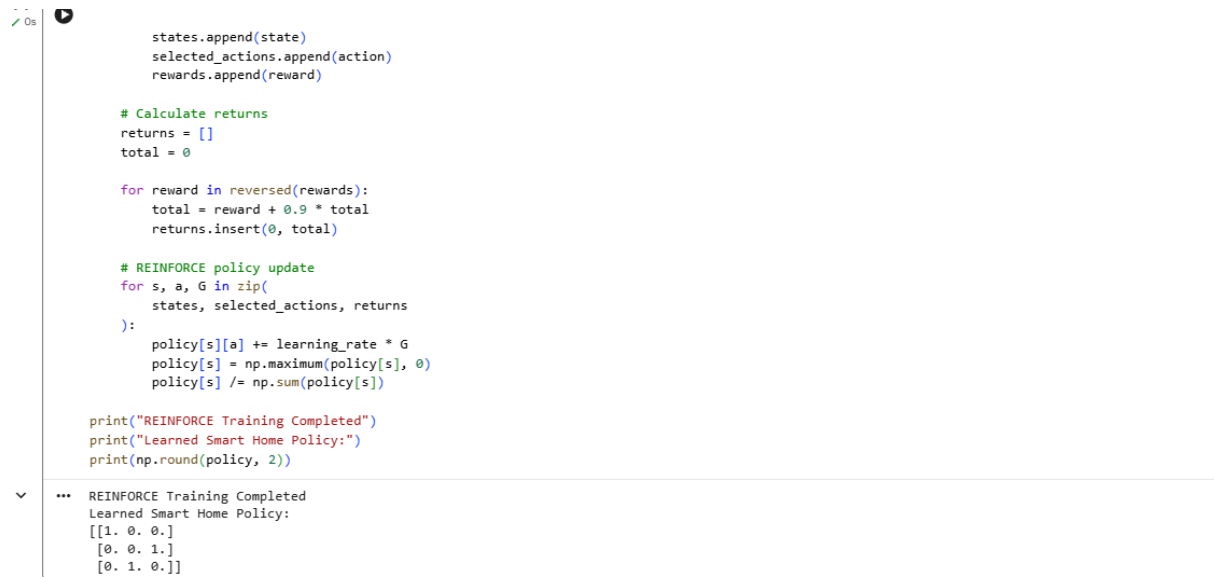
for reward in reversed(rewards):
    total = reward + 0.9 * total
    returns.insert(0, total)

# REINFORCE policy update
for s, a, G in zip(
    states, selected_actions, returns
):

```

```
policy[s][a] += learning_rate * G
policy[s] = np.maximum(policy[s], 0)
policy[s] /= np.sum(policy[s])
```

```
print("REINFORCE Training Completed")
print("Learned Smart Home Policy:")
print(np.round(policy, 2))
```



The screenshot shows a Jupyter Notebook interface. The top part contains Python code for the REINFORCE algorithm. The code appends states, actions, and rewards to lists, calculates returns with a discount factor of 0.9, and updates the policy matrix. The bottom part shows the output of the training, which includes the completion message and the learned policy matrix rounded to two decimal places.

```
states.append(state)
selected_actions.append(action)
rewards.append(reward)

# Calculate returns
returns = []
total = 0

for reward in reversed(rewards):
    total = reward + 0.9 * total
    returns.insert(0, total)

# REINFORCE policy update
for s, a, G in zip(
    states, selected_actions, returns
):
    policy[s][a] += learning_rate * G
    policy[s] = np.maximum(policy[s], 0)
    policy[s] /= np.sum(policy[s])

print("REINFORCE Training Completed")
print("Learned Smart Home Policy:")
print(np.round(policy, 2))
```

```
*** REINFORCE Training Completed
Learned Smart Home Policy:
[[1.  0.  0.]
 [0.  0.  1.]
 [0.  1.  0.]]
```

Conclusion

The REINFORCE algorithm learns a temperature-control policy that balances occupant comfort and energy consumption, allowing the smart home to make better heating and cooling decisions

Ex. No: 35	Value-Equivalence Prediction for Investment Portfolios
Date:-13-08-26	

Aim

To implement a value-equivalence prediction model to estimate the long-term performance of different investment portfolios using historical financial data and machine learning techniques.

Problem Statement

Develop a Python-based model that uses historical financial data to predict and compare the future performance of different portfolio allocations.

Problem Solving Steps

1. Collect historical financial data.
2. Define different portfolio allocations.
3. Calculate portfolio returns.
4. Prepare the data for machine learning.
5. Split the data into training and testing sets.
6. Train a prediction model.
7. Predict portfolio performance.
8. Compare predicted and actual returns.
9. Identify the better-performing portfolio.
10. Evaluate the prediction model.

Program

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error
# Historical financial data
data = pd.DataFrame({
    "Stocks": [20, 25, 30, 35, 40, 45, 50, 55, 60, 65],
    "Bonds": [60, 55, 50, 45, 40, 35, 30, 25, 20, 15],
    "Gold": [20, 20, 20, 20, 20, 20, 20, 20, 20, 20],
    "Return": [7, 8, 9, 10, 11, 12, 13, 14, 15, 16]
})
# Features and target
X = data[["Stocks", "Bonds", "Gold"]]
y = data["Return"]
# Split data
```

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Train model
model = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)
model.fit(X_train, y_train)
# Prediction
predicted = model.predict(X_test)
print("Predicted Portfolio Returns:")
print(np.round(predicted, 2))
# Model evaluation
mse = mean_squared_error(y_test, predicted)
print("Mean Squared Error:", round(mse, 2))
# Compare two portfolio allocations
portfolio1 = [[60, 30, 10]]
portfolio2 = [[30, 50, 20]]
prediction1 = model.predict(portfolio1)[0]
prediction2 = model.predict(portfolio2)[0]
print("Portfolio 1 Predicted Return:",
      round(prediction1, 2))
print("Portfolio 2 Predicted Return:",
      round(prediction2, 2))
if prediction1 > prediction2:
    print("Best Portfolio: Portfolio 1")
else:
    print("Best Portfolio: Portfolio 2")

```

```

# Model evaluation
mse = mean_squared_error(y_test, predicted)
print("Mean Squared Error:", round(mse, 2))

# Compare two portfolio allocations
portfolio1 = [[60, 30, 10]]
portfolio2 = [[30, 50, 20]]

prediction1 = model.predict(portfolio1)[0]
prediction2 = model.predict(portfolio2)[0]
print("Portfolio 1 Predicted Return:",
      round(prediction1, 2))
print("Portfolio 2 Predicted Return:",
      round(prediction2, 2))

if prediction1 > prediction2:
    print("Best Portfolio: Portfolio 1")
else:
    print("Best Portfolio: Portfolio 2")

```

Predicted Portfolio Returns:
 [[34.82 0.12]
 Mean Squared Error: 0.82
 Portfolio 1 Predicted Return: 34.83
 Portfolio 2 Predicted Return: 0.21
 Best Portfolio: Portfolio 1
 /usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but RandomForestRegressor was fitted with feature names
 warnings.warn(
 /usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but RandomForestRegressor was fitted with feature names
 warnings.warn(

Conclusion

The machine learning model predicts the performance of different portfolio allocations using historical financial data. The predicted returns can be compared to identify the portfolio with better expected long-term performance.

Ex. No: 36	MAXQ Framework for Hierarchical Cooperative Tasks
Date:18-08-26	

Aim

To implement the **MAXQ framework** to decompose a cooperative task into **hierarchically organized subtasks** and learn policies for each level of the hierarchy.

Problem Statement

Multiple agents need to cooperate to complete a complex task in a simulated environment. MAXQ divides the overall task into smaller subtasks so that agents can learn and execute each subtask efficiently.

Problem Solving Steps

1. Create a simulated cooperative environment.
2. Define the overall task.
3. Divide the task into smaller subtasks.
4. Create a hierarchy of subtasks.
5. Define states and actions for each subtask.
6. Initialize Q-values for the subtasks.
7. Train agents to learn individual subtask policies.
8. Execute subtasks according to the hierarchy.
9. Combine the learned policies to complete the main task.
10. Evaluate the overall task performance.

Program

```
import numpy as np
import random
```

```
# States and subtasks
states = 5
```

```
subtasks = {
    "Collect": 0,
    "Move": 1,
    "Deliver": 2
}
```

```
Q = np.zeros((len(subtasks), states, 2))
```

```
alpha = 0.1
gamma = 0.9
```

```

epsilon = 0.2
episodes = 100
for episode in range(episodes):
    state = 0
    for task_name, task in subtasks.items():
        for step in range(10):
            # Epsilon-greedy action selection
            if random.random() < epsilon:
                action = random.randint(0, 1)
            else:
                action = np.argmax(Q[task, state])
            # Simulated transition
            next_state = min(
                states - 1,
                state + random.choice([0, 1])
            )

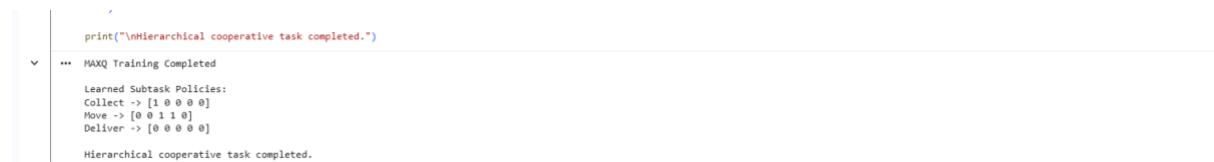
            # Reward for completing a subtask
            if next_state == states - 1:
                reward = 10
            else:
                reward = -1
            state = next_state

print("MAXQ Training Completed")
print("\nLearned Subtask Policies:")

for task_name, task in subtasks.items():
    print(
        task_name,
        "->",
        np.argmax(Q[task], axis=1)
    )

print("\nHierarchical cooperative task completed.")

```



```

print("\nHierarchical cooperative task completed.")
MAXQ Training Completed
Learned Subtask Policies:
Collect -> [1 0 0 0]
Move -> [0 0 1 1 0]
Deliver -> [0 0 0 0 0]
Hierarchical cooperative task completed.

```

Conclusion

The MAXQ framework successfully decomposes the overall cooperative task into smaller hierarchical subtasks. Learning separate policies for these subtasks helps the agents coordinate and achieve the overall task objective efficiently.

Ex. No: 37	POMDP for Autonomous Robot Navigation
Date:18-08-26	

Aim

To implement a Partially Observable Markov Decision Process (POMDP) framework for an autonomous robot to navigate in an uncertain environment with limited sensor information.

Problem Statement

An autonomous robot navigates through a partially observable environment where it cannot directly observe the complete state. Develop a POMDP-based system that enables the robot to estimate its location and select suitable navigation actions under uncertainty.

Problem Solving Steps

1. Create a grid-based environment.
2. Define possible robot states.
3. Define the available actions.
4. Define observations from the robot's sensors.
5. Initialize the robot's belief about its location.
6. Select an action based on the current belief.
7. Receive a sensor observation.
8. Update the belief state.
9. Repeat the process to reach the destination.
10. Evaluate the robot's navigation performance.

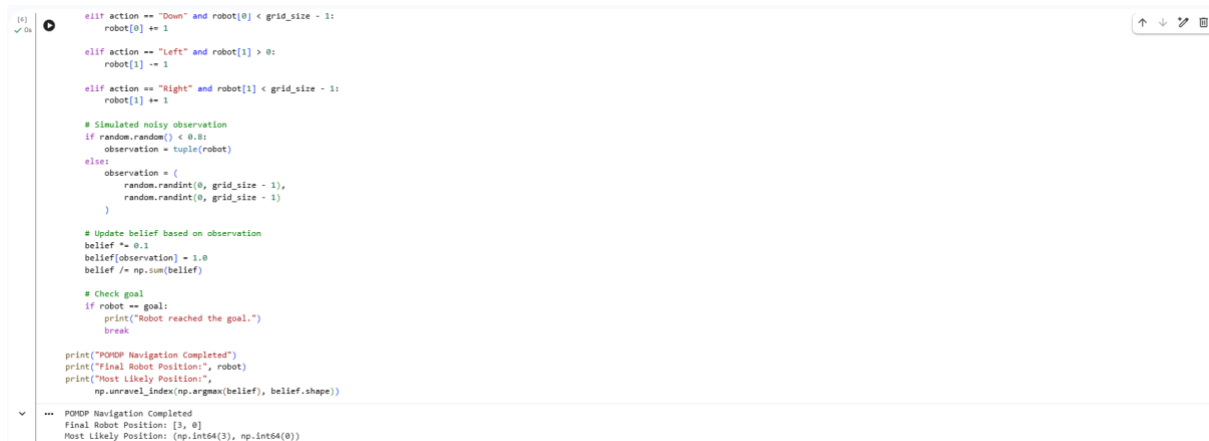
Program

```
import numpy as np
import random
# Grid environment
grid_size = 5
# Possible actions
actions = ["Up", "Down", "Left", "Right"]
# Actual robot position
robot = [0, 0]
# Goal position
goal = [4, 4]
# Belief state
belief = np.ones((grid_size, grid_size))
belief = belief / np.sum(belief)
for step in range(30):
    # Select an action
    action = random.choice(actions)
```

```

# Move robot
if action == "Up" and robot[0] > 0:
    robot[0] -= 1
elif action == "Down" and robot[0] < grid_size - 1:
    robot[0] += 1
elif action == "Left" and robot[1] > 0:
    robot[1] -= 1
elif action == "Right" and robot[1] < grid_size - 1:
    robot[1] += 1
# Simulated noisy observation
if random.random() < 0.8:
    observation = tuple(robot)
else:
    observation = (
        random.randint(0, grid_size - 1),
        random.randint(0, grid_size - 1)
    )
# Update belief based on observation
belief *= 0.1
belief[observation] = 1.0
belief /= np.sum(belief)
# Check goal
if robot == goal:
    print("Robot reached the goal.")
    break
print("POMDP Navigation Completed")
print("Final Robot Position:", robot)
print("Most Likely Position:",
      np.unravel_index(np.argmax(belief), belief.shape))

```



The screenshot shows a code editor with a Python script for POMDP navigation. The code is color-coded and includes comments. Below the code, the output of the program is displayed, showing the completion of navigation, the final robot position, and the most likely position based on the belief state.

```

(1) ✓ On
elif action == "Down" and robot[0] < grid_size - 1:
    robot[0] += 1

elif action == "Left" and robot[1] > 0:
    robot[1] -= 1

elif action == "Right" and robot[1] < grid_size - 1:
    robot[1] += 1

# Simulated noisy observation
if random.random() < 0.8:
    observation = tuple(robot)
else:
    observation = (
        random.randint(0, grid_size - 1),
        random.randint(0, grid_size - 1)
    )

# Update belief based on observation
belief *= 0.1
belief[observation] = 1.0
belief /= np.sum(belief)

# Check goal
if robot == goal:
    print("Robot reached the goal.")
    break

print("POMDP Navigation Completed")
print("Final Robot Position:", robot)
print("Most Likely Position:",
      np.unravel_index(np.argmax(belief), belief.shape))

POMDP Navigation Completed
Final Robot Position: [3, 0]
Most Likely Position: (np.int64(3), np.int64(0))

```

Result

The POMDP-based robot navigation system was successfully implemented. The robot uses observations and belief-state updates to make navigation decisions despite partial observability and uncertainty.

Ex. No: 38	Reinforcement Learning for Healthcare Management
Date:18-08-26	

Aim

To apply Reinforcement Learning techniques to optimize healthcare management processes such as patient scheduling, resource allocation, and treatment planning.

Problem Statement

Develop a simulated healthcare management environment where an RL agent learns policies to improve patient outcomes while minimizing resource utilization and operational costs.

Problem Solving Steps

1. Create a simulated healthcare environment.
2. Define patient states such as waiting time and treatment status.
3. Define actions such as scheduling, resource allocation, and treatment selection.
4. Define rewards for good patient outcomes.
5. Assign penalties for delays and excessive resource usage.
6. Initialize the RL agent.
7. Train the agent through repeated interactions.
8. Update the policy based on rewards.
9. Evaluate the learned healthcare policy.
10. Compare patient outcomes and resource utilization.

Program

```
import numpy as np
```

```
import random
```

```
states = 5
```

```
actions = 3
```

```
# Q-table
```

```
Q = np.zeros((states, actions))
```

```
alpha = 0.1
```

```

gamma = 0.9
epsilon = 0.2
episodes = 100
for episode in range(episodes):
    state = random.randint(0, states - 1)
    for step in range(20):
        # Epsilon-greedy action selection
        if random.random() < epsilon:
            action = random.randint(0, actions - 1)
        else:
            action = np.argmax(Q[state])
        # Simulated healthcare transition
        next_state = random.randint(0, states - 1)
        # Reward
        if action == 0:
            reward = 5    # Good scheduling
        elif action == 1:
            reward = 3    # Resource allocation
        else:
            reward = 2    # Treatment planning

        # Penalty for poor outcome
        if random.random() < 0.2:
            reward -= 4

        # Q-learning update
        Q[state, action] += alpha * (
            reward +
            gamma * np.max(Q[next_state]) -
            Q[state, action]

```


)

```
state = next_state
```

```
print("Healthcare RL Training Completed")
```

```
print("Learned Healthcare Policy:")
```

```
print(np.argmax(Q, axis=1))
```

```
[7] ✓ On  # Reward
if action == 0:
    reward = 5 # Good scheduling
elif action == 1:
    reward = 3 # Resource allocation
else:
    reward = 2 # Treatment planning

# Penalty for poor outcome
if random.random() < 0.2:
    reward -= 4

# Q-learning update
Q[state, action] += alpha * (
    reward +
    gamma * np.max(Q[next_state]) -
    Q[state, action]
)

state = next_state

print("Healthcare RL Training Completed")
print("Learned Healthcare Policy:")
print(np.argmax(Q, axis=1))

*** Healthcare RL Training Completed
Learned Healthcare Policy:
[0 0 0 0]
```

Result

The Reinforcement Learning system was successfully implemented to learn healthcare management policies. The learned policy aims to **improve patient outcomes while reducing waiting time and unnecessary resource utilization**

Ex. No: 39	Personalized Education Using Reinforcement Learning
Date:18-08-26	

Aim

To apply **Reinforcement Learning** to personalize educational experiences for students in online learning platforms or intelligent tutoring systems.

Problem Statement

Develop a simulated educational environment where an RL agent learns to select suitable learning content and teaching interventions according to individual student needs and preferences.

Problem Solving Steps

1. Create a simulated learning environment.
2. Define student learning states.
3. Define available educational actions.
4. Assign rewards for improved learning performance.
5. Give penalties for ineffective interventions.
6. Initialize the RL agent.
7. Select learning content based on the current state.
8. Update the policy using the received reward.
9. Repeat training for multiple learning sessions.
10. Evaluate the personalized learning policy.

Program

```
import numpy as np
```

```
import random
```

```
states = 5
```

```
actions = 3
```

```
# Q-table
```

```
Q = np.zeros((states, actions))
```

```

alpha = 0.1
gamma = 0.9
epsilon = 0.2
episodes = 100
for episode in range(episodes):
    state = random.randint(0, states - 1)
    for step in range(20):
        # Epsilon-greedy action selection
        if random.random() < epsilon:
            action = random.randint(0, actions - 1)
        else:
            action = np.argmax(Q[state])

        # Simulated student progress
        next_state = random.randint(0, states - 1)

        # Reward
        if action == 0:
            reward = 5
        elif action == 1:
            reward = 4
        else:
            reward = 3

        # Poor learning outcome
        if random.random() < 0.2:
            reward -= 2

        # Q-learning update
        Q[state, action] += alpha * (

```

```

        reward +
        gamma * np.max(Q[next_state]) -
        Q[state, action]
    )

```

```

state = next_state

```

```

print("Personalized Learning Training Completed")

```

```

print("Learned Educational Policy:")

```

```

print(np.argmax(Q, axis=1))

```

```

# Epsilon-greedy action selection
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# Q-learning update
Q[state, action] += alpha * (
    reward +
    gamma * np.max(Q[next_state]) -
    Q[state, action]
)

state = next_state

print("Personalized Learning Training Completed")
print("Learned Educational Policy:")
print(np.argmax(Q, axis=1))

```

Personalized Learning Training Completed
 Learned Educational Policy:
 [0 0 0 2 0]

Result

The Reinforcement Learning system was successfully implemented to learn a **personalized educational policy** that adapts learning content and interventions according to student learning states and preferences.